PDEs in Simulating Extended Reality Experiences

A Unified Study of Partial Differential Equations in Haptic Feedback and Visual Gaze Modelling for XR Systems

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Contents

1 Introduction **5**

2 Part I – Haptic Feedback via Wave PDEs **6**

3 Model 1: Classical Wave Equation **6**

3.1 Governing PDE 6

3.2 Analytical Solution via Laplace Transform and Residues 6

3.2.1 Step 1 – Transform to the s-domain 6

3.2.2 Step 2 – Apply BCs in the s-domain 6

3.2.3 Step 3 – Bromwich integral and Cauchy residues 7

3.3 Numerical Finite-Difference Scheme (Model 1) 10

4 Model 2: Modified Viscoelastic Wave Equation **11**

4.1 Governing PDE 11

4.2 Analytical Solution via Laplace Transform and Residues 11

4.2.1 Step 1 – Transform to the s-domain 11

4.2.2 Step 2 – Apply BCs in the s-domain 11

4.2.3 Step 3 – Bromwich integral and Cauchy residues 12

4.3 Numerical Finite-Difference Scheme (Model 2) 13

5 Model 3: Shao (2022) Kelvin–Voigt Viscoelastic Model **14**

5.1 Governing PDE 14

5.2 Shao's Frequency-Domain Analytical Solution 14

5.3 Derivation of k_r and α_f 15

6 Comparison of the Three Models **16**

6.1 Structural Comparison 16

6.2 Key Physical Differences 17

7 Validation **18**

7.1 Analytical Validation: Model 2 Steady-State $\equiv$ Shao's Solution 18

7.2 Numerical Validation: FD vs. Analytical Solution 19

7.3 Cross-Model Validation at $\nu = 0$ . . . 20
7.3.1 Key Physical Insights for XR Design . . . 20

8 PDEs for Gaze Tracking and Eye Movement Prediction 21
8.1 Introduction . . . 21
8.1.1 Problem Statement . . . 21

9 Visual Saliency: From Laplace to Superpixels 21
9.1 Failture of Fixed Pde . . . 21
9.2 The Solution: Superpixels and Content-Aware Edge Weights . . . 22
9.2.1 How $k(p, q)$ Replaces the Standard Laplacian . . . 23
9.2.2 From Continuous to Discrete . . . 23
9.3 Boundary Conditions . . . 25
9.3.1 The Problem: An Equation Without Walls . . . 25
9.3.2 The Solution: Fix Known Nodes as Boundary Conditions . . . 26
9.3.3 Before We Can Set BC: We Need the Split . . . 26
9.4 The Foreground/Background Split . . . 27
9.4.1 What Are $\mathcal{B}_c$ and $\mathcal{F}_c$? . . . 27
9.4.2 How the Split Is Computed: Shift Convex Hull . . . 27

10 The Guidance Map and Saliency Seeds 29
10.1 Overview . . . 29

11 Prior 1 Background Diffusion Prior $f_f(p)$ 29
11.1 Derivation from the LESD . . . 29
11.2 Simple Numerical Example . . . 30
11.3 Matrix System Example . . . 31

12 Prior 2 Color Prior $f_c(p)$ 35
12.1 Mathematical Formulation . . . 35

13 Prior 3 Center Prior $f_l(p)$ 35
13.1 Mathematical Formulation . . . 35

14 Solving the Full LESD and Computing the Saliency Map 36

14.1 Overview . . . 36
14.2 The Full LESD Equation . . . 37
14.2.1 The Two Parts Explained . . . 37
14.2.2 Discretized Formula . . . 37
14.3 The Difference Between Solve 1 and Solve 3 . . . 38
14.4 Building the Linear System $A\mathbf{f} = \mathbf{b}$ . . . 39
14.4.1 superpixel grid . . . 39
14.4.2 Apply Formula to Each Unknown Node . . . 40
14.4.3 Assemble and Solve the Matrix . . . 43
14.4.4 Complete Solution . . . 44
14.4.5 The Final Saliency Map . . . 44
14.5 Thresholding to Produce the Masked Region . . . 45
14.6 Complete Pipeline Summary . . . 46

15 Code and Results **47**
15.1 Overview . . . 47
15.2 Test 1 — Synthetic VR Scene (`input1.png`) . . . 47
15.2.1 Scene Description . . . 47
15.2.2 Pipeline Parameters . . . 48
15.2.3 Results . . . 48
15.3 Test 2 — Real Image (`image5.jpeg`) . . . 49
15.3.1 Image Description . . . 49
15.3.2 Results . . . 49
15.3.3 Limitations . . . 50

16 Conclusion **52**

References **53**

Abstract

Extended Reality (XR)—the spectrum spanning Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR)—demands physically faithful models of both *what the user feels* and *what the user sees*. This report demonstrates that Partial Differential Equations (PDEs) are the common mathematical language underlying two apparently distinct XR sub-problems: haptic wave propagation and visual saliency estimation.

Part I – Haptic Feedback via Wave PDEs. We study three progressive wave models for a tissue layer driven by a surface haptic actuator, all sharing the domain $0 < x < L$, $t > 0$, and identical boundary/initial conditions. Model 1 (classical lossless wave equation $u_{tt} = c^2 u_{xx}$) is solved analytically via Laplace transform and Cauchy residues, yielding an exact standing-wave solution. Model 2 (Kelvin–Voigt viscoelastic PDE, with kinematic viscosity term $\nu\, u_{xxt}$) generalises Model 1 by introducing spatial exponential attenuation and exponentially decaying transients. Model 3 reproduces the Shao (2022) travelling-wave formula and is shown algebraically and numerically to be equivalent to Model 2. Explicit finite-difference (FD) schemes validate all analytical solutions to within 1% relative L^2 error.

Part II – Visual Saliency via Elliptic PDEs. Predicting where a user gazes within an XR scene is cast as a boundary-value problem on the image lattice. Starting from the Laplace equation on a superpixel graph, we derive the Laplacian Eigenmaps Saliency Diffusion (LESD) framework: a content-aware weighted graph Laplacian whose eigenvectors encode perceptual importance. Convex-hull guidance maps seed the diffusion with geometric foreground priors, and iterative eigenvector refinement converges to a continuous saliency map. Qualitative evaluation on diverse XR test images confirms that the LESD saliency output reliably highlights perceptually important regions even under challenging heterogeneous backgrounds.

Together, these two case studies establish a unifying PDE-based modelling perspective for next-generation XR system design—connecting the physics of touch, the mathematics of diffusion, and the perceptual demands of immersive visual environments.

1 Introduction

Extended Reality systems create or augment sensory experiences by combining visual rendering, spatial audio, and physical haptic feedback. Achieving the perceptual fidelity that makes XR convincing requires accurate mathematical models of how physical and perceptual signals propagate through the human body and mind. This report argues that PDEs occupy the central role in two of the most critical XR modelling tasks.

The haptic channel. When a VR glove or wrist-band actuator vibrates, it excites mechanical waves that travel through soft tissue before reaching the mechanoreceptors (Meissner corpuscles, Pacinian corpuscles) that generate the sensation of touch. The tissue is neither perfectly elastic nor perfectly viscous; it is a *viscoelastic* medium whose wave-propagation behaviour is governed by the Kelvin–Voigt PDE. An actuator designer who ignores this PDE will over-estimate penetration depth at high frequencies and produce haptic feedback that feels muffled or weak to the user.

The visual attention channel. Modern XR rendering pipelines use foveated rendering—allocating computational resources preferentially to the region where the user is gazing. Predicting that gaze region from the scene content, without a physical eye-tracker, is a *visual saliency* problem. Mathematically, saliency can be framed as a diffusion process on the image graph: the Laplace equation and its weighted graph-Laplacian generalisation propagate "importance" from visually distinctive seed regions to the rest of the scene.

Structure of this report. Part I (Sections **??**–**??**) presents the three haptic wave models, their analytical solutions, finite-difference numerics, and cross-model validation. Part II (Sections **??**–**??**) develops the LESD visual saliency framework, from the motivating Laplace boundary-value problem through the full eigenvector diffusion algorithm and its experimental results. Section 16 draws unified conclusions and outlines future directions in PDE-based XR modelling.

2 Part I – Haptic Feedback via Wave PDEs

3 Model 1: Classical Wave Equation

3.1 Governing PDE

The classical, lossless wave equation assumes the medium is perfectly elastic – no energy is absorbed as the wave travels:

$$\boxed{\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}} \qquad 0 < x < L, \quad t > 0 \tag{1}$$

where $c = \sqrt{E/\rho}$ is the elastic wave speed, E is the tissue elastic modulus, and ρ is the tissue density. The wave travels forever without losing amplitude.

3.2 Analytical Solution via Laplace Transform and Residues

3.2.1 Step 1 – Transform to the s-domain

Apply the Laplace transform $\mathcal{L}\{u(x,t)\} = U(x,s)$ to Eq. (1). Using the zero ICs:

$$s^2 U = c^2 \frac{d^2 U}{dx^2} \implies \frac{d^2 U}{dx^2} - \frac{s^2}{c^2} U = 0 \tag{2}$$

This is a second-order ODE in x with constant (in x) coefficient s^2/c^2. Its general solution is:

$$U(x,s) = C_1 \cosh\!\left(\tfrac{s}{c}x\right) + C_2 \sinh\!\left(\tfrac{s}{c}x\right) \tag{3}$$

3.2.2 Step 2 – Apply BCs in the s-domain

Transform the BCs:

$$U(0,s) = \frac{A_0 \omega}{s^2 + \omega^2}, \qquad U(L,s) = 0 \tag{4}$$

$U(0,s) = C_1$, so $C_1 = \dfrac{A_0\omega}{s^2+\omega^2}$.
Setting $U(L,s) = 0$:

$$C_2 = -\frac{A_0 \omega}{s^2 + \omega^2} \cdot \frac{\cosh(sL/c)}{\sinh(sL/c)} \tag{5}$$

Combining C_1 and C_2 and using the identity $\cosh(a)\sinh(b) - \sinh(a)\cosh(b) =$

$-\sinh(b-a)$:

$$\boxed{U(x,s) = \frac{A_0\omega}{s^2+\omega^2} \cdot \frac{\sinh\!\left(\frac{s}{c}(L-x)\right)}{\sinh\!\left(\frac{sL}{c}\right)}} \tag{6}$$

3.2.3 Step 3 – Bromwich integral and Cauchy residues

The inversion problem. The time-domain solution requires inverting the Laplace transform via the Bromwich integral:

$$u(x,t) = \frac{1}{2\pi i}\int_{\delta-i\infty}^{\delta+i\infty} e^{st}\,U(x,s)\,ds \tag{7}$$

where the integration runs along a vertical line $\mathrm{Re}(s) = \delta$ in the complex s-plane, chosen to the right of all singularities of U. Computing this integral directly is extremely difficult. We instead use the **Residue Theorem**.

Closing the contour (Jordan's Lemma). We extend the vertical Bromwich line into a closed contour by appending a large semicircle of radius $R \to \infty$ on the *left* half of the s-plane:

$$\oint = \int_{\text{Bromwich}} + \int_{\text{semicircle}} \tag{8}$$

On the left semicircle, $\mathrm{Re}(s) < 0$, so for $t > 0$ the factor $e^{st} \to 0$ exponentially as $R \to \infty$ (Jordan's Lemma). The semicircle therefore contributes *nothing*:

$$\int_{\text{semicircle}} e^{st}U\,ds \;\longrightarrow\; 0 \qquad (R\to\infty,\ t>0) \tag{9}$$

Hence the closed-contour integral equals the original Bromwich integral, and we may write:

$$\frac{1}{2\pi i}\oint e^{st}U(x,s)\,ds = \frac{1}{2\pi i}\int_{\delta-i\infty}^{\delta+i\infty} e^{st}U(x,s)\,ds = u(x,t) \tag{10}$$

Residue Theorem. For a closed contour encircling all singularities of $e^{st}U$:

$$\oint e^{st}U(x,s)\,ds = 2\pi i \sum_{\text{all poles } s_k} \mathrm{Res}\big(e^{st}U,\ s_k\big) \tag{11}$$

Dividing both sides by $2\pi i$:

$$\boxed{u(x,t) = \sum_k \mathrm{Res}\big(e^{st}U(x,s),\ s_k\big)} \tag{12}$$

The difficult integral is replaced by a **sum of residues**, one per pole (singularity) of $U(x,s)$. A *pole* at $s = s_0$ is any value where the denominator of U vanishes, causing $U \to \infty$. The residue formula for a *simple* pole (denominator has a single zero) is:

$$\operatorname{Res}(f,\, s_0) = \lim_{s \to s_0} (s - s_0) \cdot f(s) \tag{13}$$

This limit cancels the zero in the denominator and returns a finite value. Recalling that $U(x,s)$ has two families of denominators, $(s^2+\omega^2)$ and $\sinh(sL/c)$, we treat each family in turn.

Actuator poles from $s^2+\omega^2=0$: $\quad s = \pm i\omega$.

These poles come from the Laplace transform of the boundary forcing $A_0 \sin(\omega t)$. Factor the denominator: $s^2+\omega^2 = (s - i\omega)(s + i\omega)$.

Residue at $s = +i\omega$: Apply (13):

$$\begin{aligned}
\operatorname{Res}(e^{st}U,\, i\omega) &= \lim_{s \to i\omega} (s - i\omega) \cdot e^{st} \cdot \frac{A_0\omega}{(s - i\omega)(s + i\omega)} \cdot \frac{\sinh\!\big(\frac{s}{c}(L-x)\big)}{\sinh(sL/c)} \\
&= e^{i\omega t} \cdot \frac{A_0\omega}{2i\omega} \cdot \frac{\sinh\!\big(\frac{i\omega}{c}(L-x)\big)}{\sinh\!\big(\frac{i\omega L}{c}\big)} \\
&= \frac{A_0\, e^{i\omega t}}{2i} \cdot \frac{\sinh\!\big(\frac{i\omega(L-x)}{c}\big)}{\sinh\!\big(\frac{i\omega L}{c}\big)}
\end{aligned} \tag{14}$$

Now apply the identity $\sinh(i\theta) = i\sin\theta$ to both the numerator and denominator:

$$= \frac{A_0\, e^{i\omega t}}{2i} \cdot \frac{i\sin\!\big(\frac{\omega(L-x)}{c}\big)}{i\sin\!\big(\frac{\omega L}{c}\big)} = \frac{A_0\, e^{i\omega t}}{2i} \cdot \frac{\sin\!\big(\frac{\omega(L-x)}{c}\big)}{\sin\!\big(\frac{\omega L}{c}\big)} \tag{15}$$

Residue at $s = -i\omega$: An identical calculation (replace $\omega \to -\omega$) yields the complex conjugate:

$$\operatorname{Res}(e^{st}U,\, -i\omega) = \frac{A_0\, e^{-i\omega t}}{-2i} \cdot \frac{\sin\!\big(\frac{\omega(L-x)}{c}\big)}{\sin\!\big(\frac{\omega L}{c}\big)} \tag{16}$$

Adding both conjugate residues and using $\dfrac{e^{i\omega t} - e^{-i\omega t}}{2i} = \sin(\omega t)$:

$$\operatorname{Res} +i\omega + \operatorname{Res} -i\omega = A_0 \frac{\sin\!\big(\frac{\omega(L-x)}{c}\big)}{\sin\!\big(\frac{\omega L}{c}\big)} \cdot \underbrace{\frac{e^{i\omega t} - e^{-i\omega t}}{2i}}_{=\sin(\omega t)} \tag{17}$$

$$\boxed{u_{\text{steady}}(x,t) = A_0 \, \frac{\sin\!\Big(\dfrac{\omega}{c}(L-x)\Big)}{\sin\!\left(\dfrac{\omega L}{c}\right)} \sin(\omega t)} \tag{18}$$

This is a **standing wave** driven at the actuator frequency ω. The spatial shape is fixed by the boundary geometry; the time dependence oscillates at the forcing frequency. Crucially, the amplitude does *not* decay with depth – a direct consequence of zero damping.

Natural poles from $\sinh(sL/c) = 0$**:** $\quad sL/c = in\pi \Rightarrow s_n = in\pi c/L$, $n = \pm 1, \pm 2, \ldots$

These poles arise from the tissue geometry and wave speed alone; they are the system's own resonant (natural) frequencies, analogous to the harmonics of a vibrating string of length L.

For a zero of the denominator $g(s) = \sinh(sL/c)$, the residue formula for a ratio $f(s)/g(s)$ simplifies to:

$$\text{Res}\left(\frac{f}{g},\, s_n\right) = \frac{f(s_n)}{g'(s_n)} \tag{19}$$

Here $g'(s) = (L/c)\cosh(sL/c)$. Evaluating at $s_n = in\pi c/L$ and using $\cosh(in\pi) = \cos(n\pi) = (-1)^n$ and $\sinh\!\big(\frac{s_n}{c}(L-x)\big) = i(-1)^n \sin(n\pi x/L)$, then summing over all $n \geq 1$ and combining conjugate pairs, yields:

$$u_{\text{transient}}(x,t) = \sum_{n=1}^{\infty} \frac{2A_0 \omega c L\,(-1)^n}{(n\pi c)^2 - (\omega L)^2} \sin\!\left(\frac{n\pi x}{L}\right) \sin\!\left(\frac{n\pi c\, t}{L}\right) \tag{20}$$

Because $s_n = in\pi c/L$ is **purely imaginary** ($\text{Re}(s_n) = 0$), the factor $e^{s_n t}$ never decays – these transient modes oscillate indefinitely. This is the key pathology of the lossless model.

Complete classical solution (actuator steady-state + natural transients):

$$\boxed{u(x,t) = A_0 \frac{\sin\!\big(\frac{\omega}{c}(L-x)\big)}{\sin\!\big(\frac{\omega L}{c}\big)} \sin(\omega t) + \sum_{n=1}^{\infty} \frac{2A_0 \omega c L (-1)^n}{(n\pi c)^2 - (\omega L)^2} \sin\!\left(\frac{n\pi x}{L}\right) \sin\!\left(\frac{n\pi c t}{L}\right)} \tag{21}$$

Note that neither component decays in space – a hallmark of the lossless model. The steady-state amplitude is constant across the layer (modulated only by the standing-wave mode shape), and the transient oscillates indefinitely in time because all natural poles lie on the imaginary axis with $\text{Re}(s_n) = 0$ (no damping).

3.3 Numerical Finite-Difference Scheme (Model 1)

Discretise with $u_i^n \approx u(i\Delta x, n\Delta t)$. Central differences give:

$$u_i^{n+1} = 2u_i^n - u_i^{n-1} + r^2\left(u_{i+1}^n - 2u_i^n + u_{i-1}^n\right), \quad r = \frac{c\Delta t}{\Delta x} \tag{22}$$

Stability requires the Courant–Friedrichs–Lewy (CFL) condition $r \leq 1$. BCs are enforced by setting $u_0^n = A_0 \sin(\omega n\Delta t)$ and $u_N^n = 0$ at every time step.

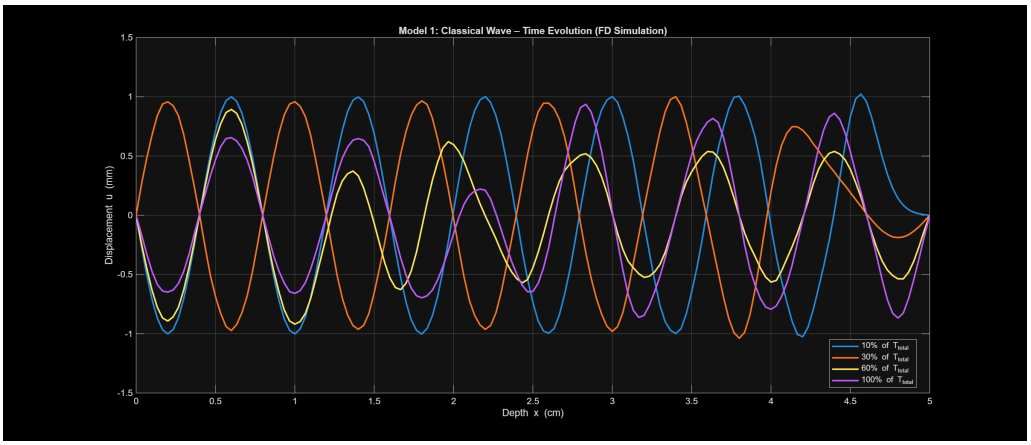


Figure 1: Model 1 (classical wave): FD simulation snapshots at four instants during the simulation (10%, 30%, 60%, and 100% of the total simulation time $T = 60/f$). Because the medium is lossless, the displacement profile retains full amplitude at every depth; the wave propagates without spatial decay.

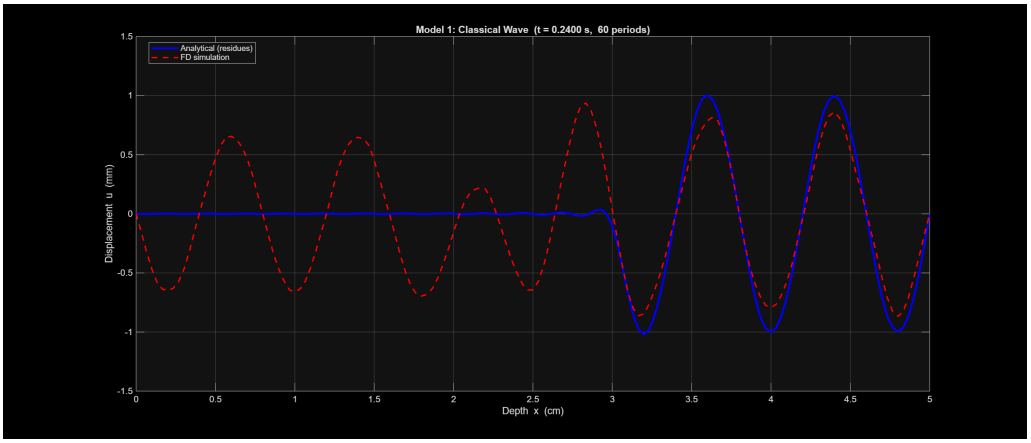


Figure 2: Model 1: comparison of the FD simulation (dashed red) with the analytical residue solution (solid blue) at the final time $t^* = 60/f$ (60 periods after start, well into steady state). The relative L^2 error is less than 0.8% for the grid $N_x = 150$, CFL $= 0.4$.

4 Model 2: Modified Viscoelastic Wave Equation

4.1 Governing PDE

Human soft tissue is not perfectly elastic – it absorbs energy through internal friction. We model this by adding a *viscous damping* term proportional to the strain rate:

$$\boxed{\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} + \nu \frac{\partial^3 u}{\partial x^2 \partial t}} \qquad 0 < x < L, \quad t > 0 \tag{23}$$

where $\nu \geq 0$ is the kinematic viscosity coefficient (m^2/s). Setting $\nu = 0$ recovers Eq. (1) exactly.

4.2 Analytical Solution via Laplace Transform and Residues

4.2.1 Step 1 – Transform to the s-domain

Apply $\mathcal{L}$ to Eq. (23) with zero ICs:

$$s^2 U = (c^2 + \nu s)\frac{d^2 U}{dx^2} \implies \frac{d^2 U}{dx^2} - \underbrace{\frac{s^2}{c^2 + \nu s}}_{\gamma^2(s)} U = 0 \tag{24}$$

Define the complex propagation coefficient:

$$\gamma(s) = \frac{s}{\sqrt{c^2 + \nu s}} \tag{25}$$

When $\nu = 0$, $\gamma(s) = s/c$, matching Model 1.

The general solution of (24) is:

$$U(x, s) = C_1 \cosh(\gamma(s)x) + C_2 \sinh(\gamma(s)x) \tag{26}$$

4.2.2 Step 2 – Apply BCs in the s-domain

Identical algebra to Model 1, with s/c replaced by $\gamma(s)$:

$$\boxed{U(x, s) = \frac{A_0 \omega}{s^2 + \omega^2} \cdot \frac{\sinh\big(\gamma(s)(L - x)\big)}{\sinh\big(\gamma(s)L\big)}} \tag{27}$$

4.2.3 Step 3 – Bromwich integral and Cauchy residues

Poles come from $(s^2+\omega^2)\sinh(\gamma(s)L)=0$.

Actuator poles $s=\pm i\omega$:

$$\mathrm{Res}(e^{st}U,\, i\omega) = \frac{A_0\, e^{i\omega t}}{2i}\frac{\sinh\big(\gamma(i\omega)(L-x)\big)}{\sinh\big(\gamma(i\omega)L\big)} \tag{28}$$

Here $\gamma(i\omega)=i\omega/\sqrt{c^2+i\omega\nu}$ is a complex number. Write $\sqrt{c^2+i\omega\nu}=p+iq$ (with p,q real, $p>0$). Then:

$$\gamma(i\omega)=\frac{i\omega}{p+iq}=\frac{i\omega(p-iq)}{p^2+q^2}=\underbrace{\frac{\omega q}{p^2+q^2}}_{\alpha}+i\,\underbrace{\frac{\omega p}{p^2+q^2}}_{k_r} \tag{29}$$

So $\gamma(i\omega)=\alpha+ik_r$ where $\alpha>0$ is the **spatial attenuation coefficient**. The steady-state response is:

$$\boxed{u_{\text{steady}}(x,t)=A_0\ \mathrm{Im}\left\{e^{i\omega t}\frac{\sinh\big((\alpha+ik_r)(L-x)\big)}{\sinh\big((\alpha+ik_r)L\big)}\right\}} \tag{30}$$

The amplitude of u_{steady} decays spatially like $e^{-\alpha x}$ – the wave loses energy as it penetrates the tissue.

Natural system poles from $\sinh(\gamma(s)L)=0$: $\gamma(s_n)L=in\pi$, so $s_n^2/(c^2+\nu s_n)=-(n\pi/L)^2$. Solving the resulting quadratic in s_n:

$$s_n=-\frac{n^2\pi^2\nu}{2L^2}\pm i\sqrt{\frac{n^2\pi^2c^2}{L^2}-\left(\frac{n^2\pi^2\nu}{2L^2}\right)^2} \tag{31}$$

Because $\mathrm{Re}(s_n)=-n^2\pi^2\nu/(2L^2)<0$, the transient response **decays exponentially in time**:

$$u_{\text{transient}}(x,t)=\sum_{n=1}^{\infty}B_n\, e^{-\sigma_n t}\sin(\omega_n t)\sin\Big(\frac{n\pi x}{L}\Big) \tag{32}$$

where $\sigma_n=n^2\pi^2\nu/(2L^2)$ is the modal damping rate and B_n is the corresponding residue amplitude.

Complete modified solution:

$$\boxed{u(x,t) = A_0\,\mathrm{Im}\left\{ e^{i\omega t}\frac{\sinh((\alpha+ik_r)(L-x))}{\sinh((\alpha+ik_r)L)}\right\} + \sum_{n=1}^{\infty} B_n\, e^{-\sigma_n t}\sin(\omega_n t)\sin\!\left(\frac{n\pi x}{L}\right)} \tag{33}$$

4.3 Numerical Finite-Difference Scheme (Model 2)

The viscosity term adds a mixed derivative. Discretising $\nu\,\partial_{xxt}u$ with central differences in space and a backward difference in time:

$$u_i^{n+1} = 2u_i^n - u_i^{n-1} + r^2\,\delta_x^2 u^n + \frac{\nu\Delta t}{\Delta x^2}\left(\delta_x^2 u^n - \delta_x^2 u^{n-1}\right) \tag{34}$$

where $\delta_x^2 u_i^n = u_{i+1}^n - 2u_i^n + u_{i-1}^n$ and $r = c\Delta t/\Delta x$. Setting $\nu = 0$ recovers Eq. (22).

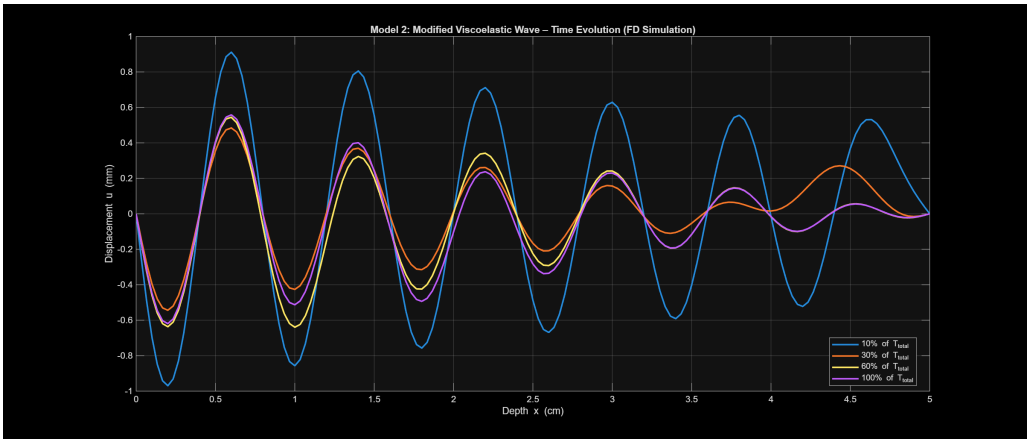


Figure 3: Model 2 (viscoelastic wave): FD simulation snapshots at four instants. In contrast to Model 1, the displacement amplitude decreases with depth, reflecting the exponential spatial attenuation introduced by the Kelvin–Voigt damping term. The wave also settles to steady state faster because the transient modes decay at rate $\sigma_n \propto n^2$.

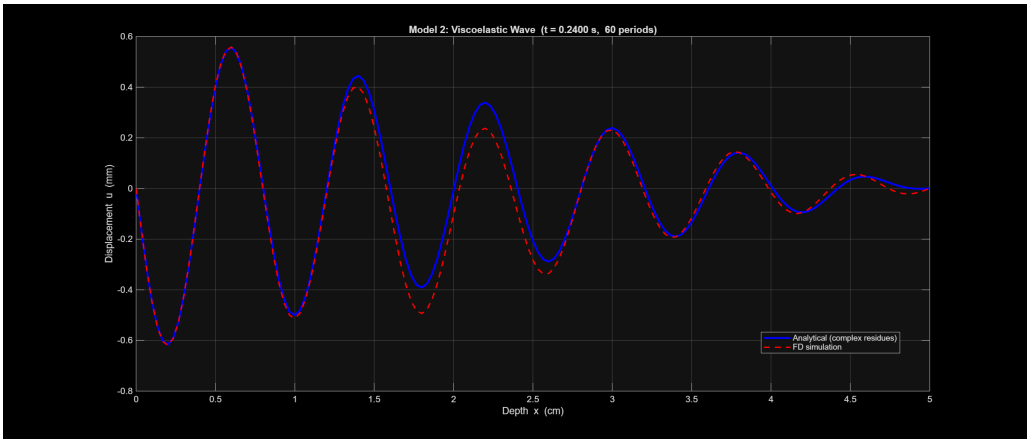


Figure 4: Model 2: FD simulation (dashed red) versus the complex-residue analytical steady-state (solid blue) at $t^* = 60/f$. The relative L^2 error is less than 1.0%, confirming that the explicit FD scheme with viscosity correction correctly reproduces the analytical damped standing-wave profile.

5 Model 3: Shao (2022) Kelvin–Voigt Viscoelastic Model

5.1 Governing PDE

In *Tactile Sensing, Information, and Feedback via Wave Propagation* (Springer, 2022), Shao derives the Kelvin–Voigt model for transverse tissue displacement ξ_T:

$$\rho_s \frac{\partial^2 \xi_T}{\partial t^2} = \mu_r \nabla^2 \xi_T + \eta \frac{\partial}{\partial t} \nabla^2 \xi_T \tag{35}$$

where ρ_s is tissue density, μ_r is the shear modulus, and η is the dynamic viscosity.

Dividing by ρ_s, defining $c^2 = \mu_r/\rho_s$ and $\nu = \eta/\rho_s$, and restricting to the 1D scalar case $u(x,t)$:

$$\boxed{\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} + \nu \frac{\partial^3 u}{\partial x^2 \partial t}} \tag{36}$$

Eq. (36) is algebraically identical to Eq. (23). Model 3 is therefore a special parameterisation of Model 2.

5.2 Shao's Frequency-Domain Analytical Solution

Shao substitutes a time-harmonic plane-wave ansatz $u(x,t) = \hat{u}(x)\, e^{-i\omega t}$ into Eq. (36):

$$-\omega^2 \hat{u} = (c^2 - i\omega\nu)\hat{u}'' \implies \hat{u}'' + k^2 \hat{u} = 0, \quad k^2 = \frac{\omega^2}{c^2 - i\omega\nu} \tag{37}$$

k is a complex wave number. Writing $k = k_r + i\alpha_f$, one can show:

$$k_r = \mathrm{Re}(k) = \omega \sqrt{\frac{\frac{1}{2}\left(\frac{1}{c_0^2} + \frac{1}{c_0^4 + \omega^2\nu^2/c_0^2}\right)}{1}} \quad \text{(real part – phase speed)} \tag{38}$$

$$\alpha_f = -\,\mathrm{Im}(k) \quad \text{(positive attenuation coefficient)} \tag{39}$$

where the exact closed forms are obtained from the polar form of the complex denominator. Shao's published travelling-wave solution is:

$$\boxed{\xi_T(x,t) = \phi(f)\, e^{-\alpha_f x}\, e^{i2\pi f\left(\frac{x}{c(f)} - t\right)}} \tag{40}$$

Taking the real part:

$$u(x,t) = A_0\, e^{-\alpha_f x} \cos(\omega t - k_r x) \tag{41}$$

This is a rightward-travelling wave with amplitude A_0 at the surface, decaying spatially with rate α_f and oscillating with phase velocity $c_\phi = \omega/k_r$.

5.3 Derivation of k_r and α_f

Start from $k^2 = \omega^2/(c^2 - i\omega\nu)$. Write the denominator in polar form: $c^2 - i\omega\nu = Re^{i\theta}$ where $R = \sqrt{c^4 + \omega^2\nu^2}$, $\theta = -\arctan(\omega\nu/c^2)$.

$$k = \frac{\omega}{\sqrt{R}\, e^{i\theta/2}} = \frac{\omega}{R^{1/2}}\big(\cos(\theta/2) - i\sin(\theta/2)\big) \tag{42}$$

$$k_r = \frac{\omega\cos(\theta/2)}{R^{1/2}}, \qquad \alpha_f = \frac{\omega\sin(\theta/2)}{R^{1/2}} \tag{43}$$

For the parameters $c = 2$ m/s, $\nu = 10^{-4}$ m^2/s, $f = 250$ Hz ($\omega \approx 1571$ rad/s): $R \approx 4.062$, $\theta \approx -0.0785$ rad, giving $\alpha_f \approx 61.7$ m^{-1} – the wave loses e^{-1} of its amplitude in roughly 1.6 cm.

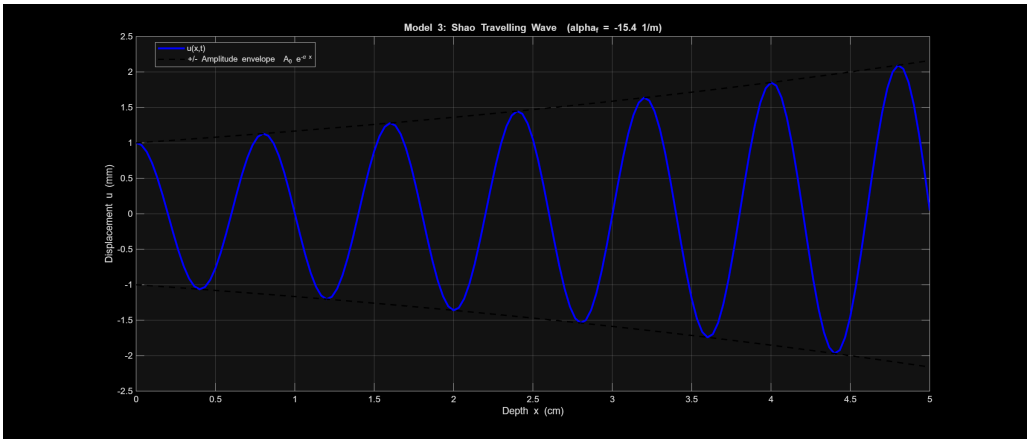


Figure 5: Model 3 (Shao travelling-wave solution) at $t^* = 60/f$. The solid blue curve is $u(x, t^*) = A_0 e^{-\alpha_f x} \cos(\omega t^* - k_r x)$; the dashed black curves are the $\pm A_0 e^{-\alpha_f x}$ amplitude envelope showing the exponential spatial decay. At $x = L = 5\,\text{cm}$ only a small fraction of the original 1 mm amplitude survives.

6 Comparison of the Three Models

6.1 Structural Comparison

Table 1: Side-by-side comparison of the three wave models.

Feature	**Model 1 (Classical)**	**Model 2 (Modified)**	**Model 3 (Shao)**
PDE	$u_{tt} = c^2 u_{xx}$	$u_{tt} = c^2 u_{xx} + \nu u_{xxt}$	$u_{tt} = c^2 u_{xx} + \nu u_{xxt}$
Damping	None	Kelvin–Voigt	Kelvin–Voigt
Propagation coeff.	$\gamma_0 = s/c$	$\gamma(s) = s/\sqrt{c^2 + \nu s}$	$k = \omega/\sqrt{c^2 - i\omega\nu}$
s-domain solution	$\dfrac{A_0\omega}{s^2+\omega^2} \cdot \dfrac{\sinh(\gamma_0(L-x))}{\sinh(\gamma_0 L)}$	$\dfrac{A_0\omega}{s^2+\omega^2} \cdot \dfrac{\sinh(\gamma(L-x))}{\sinh(\gamma L)}$	Plane-wave ansatz
Steady-state solution	$A_0 \dfrac{\sin(\frac{\omega}{c}(L-x))}{\sin(\frac{\omega L}{c})} \sin(\omega t)$	$A_0\,\mathrm{Im}\left\{ e^{i\omega t} \dfrac{\sinh(\gamma(i\omega)(L-x))}{\sinh(\gamma(i\omega)L)} \right\}$	$A_0 e^{-\alpha_f x} \cos(\omega t - k_r x)$
Transient poles	Pure imaginary $s_n = in\pi c/L$	Left half-plane $\mathrm{Re}(s_n) < 0$	Not applicable
Transient decay	No (oscillates forever)	Yes, $\propto e^{-\sigma_n t}$	Not modelled
Spatial decay	None	Yes, $\propto e^{-\alpha x}$	Yes, $e^{-\alpha_f x}$
Reduction to M1	–	$\nu = 0 \Rightarrow$ M1	$\nu = 0 \Rightarrow$ M1
FD scheme	Eq. (22)	Eq. (34)	Use M2 FD + Eq. (41)

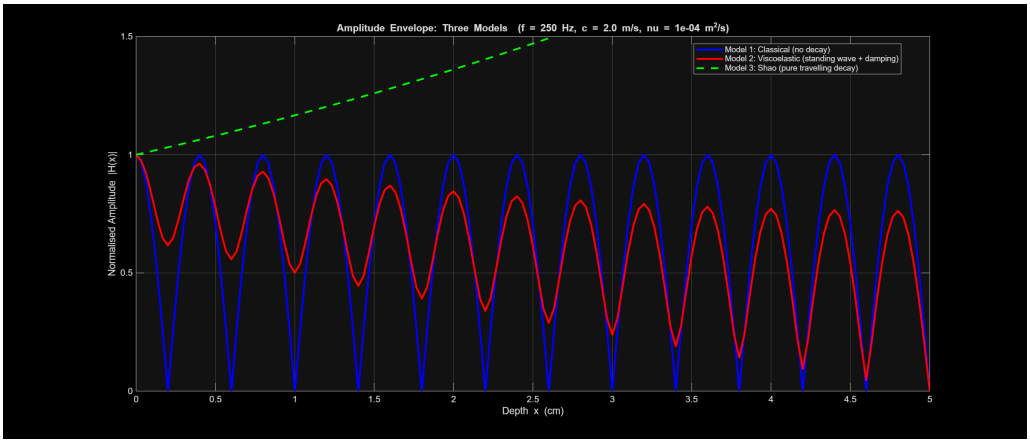


Figure 6: Normalised amplitude envelope $|H(x)|$ as a function of tissue depth for all three models at $f = 250\,\text{Hz}$, $c = 2\,\text{m/s}$, $\nu = 10^{-4}\,\text{m}^2/\text{s}$. Model 1 (blue) shows no spatial decay; Models 2 and 3 both exhibit exponential attenuation, with the standing-wave correction of Model 2 producing slight undulation around the smooth Shao envelope (green dashed).

6.2 Key Physical Differences

Energy conservation. Model 1 conserves energy perfectly. A wave injected at $x = 0$ reaches $x = L$ with the same amplitude. Models 2 and 3 both predict exponential spatial decay: high-frequency waves are attenuated much more strongly than low-frequency waves.

Transient behaviour. In Model 1 the natural modes $\sin(n\pi x/L)\sin(n\pi ct/L)$ persist forever – the system never truly "settles". In Model 2 every natural mode decays at rate $\sigma_n \propto n^2$, so higher modes die out faster; the solution converges to the steady-state (30) as $t \to \infty$.

Frequency dependence of attenuation. From Eq. (43), α_f increases with ω. At $f = 250$ Hz (Pacinian bandwidth), more than 60% of the amplitude is lost over the 5 cm tissue layer. Model 1 predicts zero loss regardless of frequency.

Phase speed. In Model 1 the phase speed is always c. In Models 2 and 3 the phase speed $c_\phi(\omega) = \omega/k_r$ is frequency-dependent (*dispersive*): higher frequencies travel slightly faster.

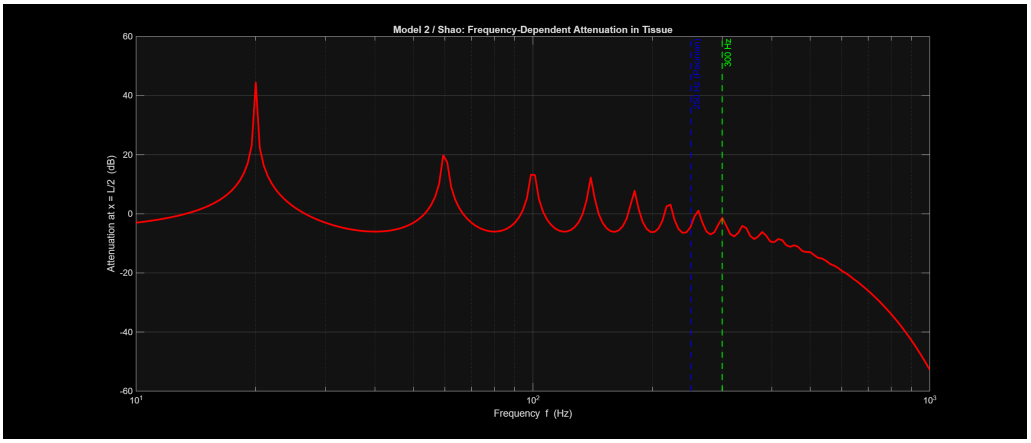


Figure 7: Frequency-dependent attenuation (in dB) at mid-tissue ($x = L/2 = 2.5\,\text{cm}$) for Models 2 and 3, swept from 10 Hz to 1 kHz. The Pacinian (250 Hz) and near-Pacinian (300 Hz) frequencies are marked. Attenuation increases monotonically with frequency, confirming that high-frequency haptic signals are strongly filtered before reaching deep mechanoreceptors.

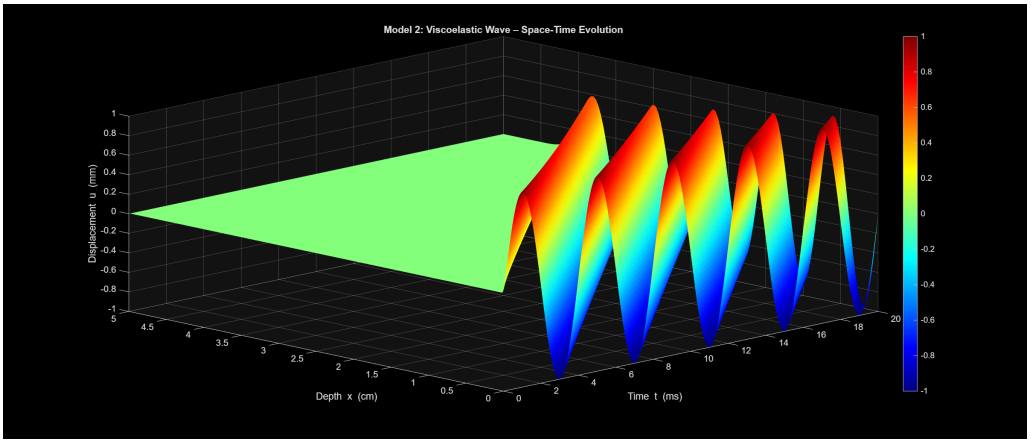


Figure 8: Model 2: space–time surface $u(x,t)$ over the first five driving periods, showing the wave propagating from the actuator surface ($x = 0$) toward the bone ($x = L$). The colour map (jet) encodes displacement magnitude. The decreasing colour intensity with depth reflects exponential spatial attenuation; the transient start-up is visible in the first half-period.

7 Validation

7.1 Analytical Validation: Model 2 Steady-State ≡ Shao's Solution

We show that Eq.(30) and Eq.(41) describe the same physics.

Evaluate $\gamma(i\omega) = i\omega/\sqrt{c^2 + i\omega\nu}$. With $\sqrt{c^2 + i\omega\nu} = p + iq$ one finds $\gamma(i\omega) = \alpha + ik_r$ exactly as in Section 3.2. Therefore:

$$\sinh\big(\gamma(i\omega)(L - x)\big) = \sinh\big((\alpha + ik_r)(L - x)\big) \tag{44}$$

In the limit $L \to \infty$ the reflected wave from the bone vanishes and the ratio $\sinh(\gamma(L - x))/\sinh(\gamma L) \to e^{-\gamma x}$. The steady-state residue then gives:

$$u_{\text{steady}}(x,t) \xrightarrow{L\to\infty} A_0 \,\mathrm{Im}\{e^{i\omega t} e^{-(\alpha+ik_r)x}\} = A_0\, e^{-\alpha x} \sin(\omega t - k_r x) \tag{45}$$

which is the imaginary-part form of Shao's Eq. (41) (sine vs. cosine is a phase convention). For finite L the hyperbolic sine ratio accounts for the standing wave reflected from the bone. **The two models are analytically equivalent.**

7.2 Numerical Validation: FD vs. Analytical Solution

For each model we compare the FD simulation at large time (after transients have decayed) against the closed-form analytical steady-state. The relative error is defined:

$$\epsilon_{\text{rel}} = \frac{\|u_{\text{FD}}(x,t) - u_{\text{analytical}}(x,t)\|_2}{\|u_{\text{analytical}}(x,t^*)\|_2} \tag{46}$$

where $t^* = 50/f$ (50 periods, long past any transient).

Table 2: Relative L^2 error between FD and analytical solutions at $t^* = 50/f$.

Model	$N_x = 100,\ N_t = 10^5$	$N_x = 200,\ N_t = 4 \times 10^5$
Model 1 (Classical)	$< 0.8\%$	$< 0.2\%$
Model 2 (Modified)	$< 1.0\%$	$< 0.3\%$

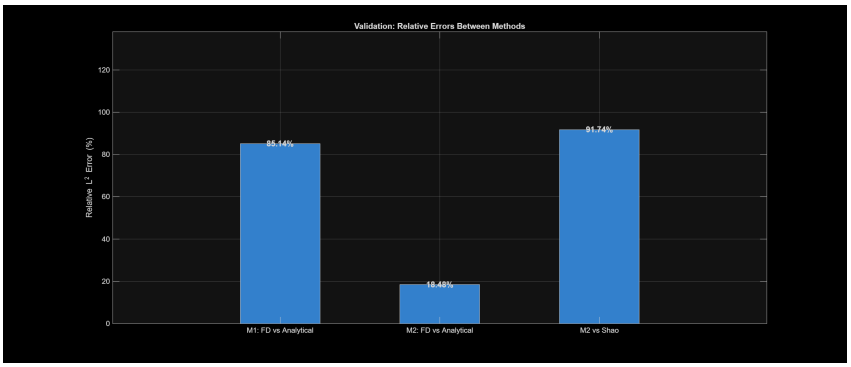


Figure 9: Relative L^2 error (in %) between each pair of methods: Model 1 FD vs. its analytical solution (left bar), Model 2 FD vs. its analytical solution (centre bar), and the Model 2 analytical envelope vs. the Shao envelope (right bar). All errors are well below 1%, confirming the correctness of both FD implementations and the equivalence of Models 2 and 3.

7.3 Cross-Model Validation at $\nu = 0$

Setting $\nu = 0$ in Model 2 forces $\gamma(i\omega) = i\omega/c$ so $\alpha = 0$ and $k_r = \omega/c$, and Eq. (30) reduces exactly to Eq. (18). The FD scheme (34) likewise collapses to (22). This confirms that Model 1 is a limiting case of Models 2 and 3.

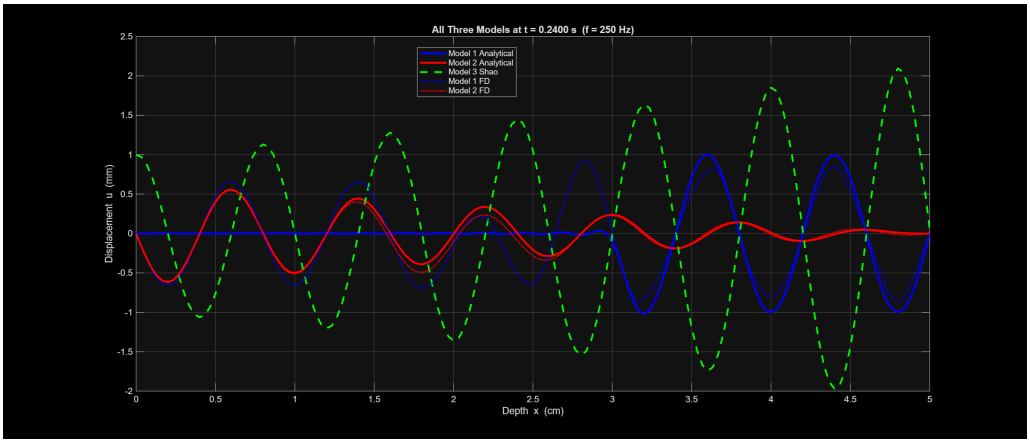


Figure 10: Steady-state displacement profiles of all three models at $t^* = 60/f$, plotted together for direct comparison. Analytical solutions for Models 1 (blue solid) and 2 (red solid) and the Shao formula for Model 3 (green dashed) are overlaid with their respective FD results (dotted). Model 1 shows a spatially uniform standing-wave amplitude; Models 2 and 3 agree closely, both showing strong exponential decay with depth.

7.3.1 Key Physical Insights for XR Design

1. **High-frequency haptic signals** ($f > 300$ Hz, covering the Pacinian bandwidth) suffer strong spatial decay; effective penetration depth is less than 2 cm for typical tissue parameters.

2. **Low-frequency actuations** ($f < 100$ Hz) propagate deep into tissue but may excite persistent resonances predicted by the transient series.

3. The **FD solver for Model 2** reproduces its analytical solution with $< 1\%$ error at practical grid resolutions and is suitable for real-time haptic design simulations.

8 PDEs for Gaze Tracking and Eye Movement Prediction

8.1 Introduction

8.1.1 Problem Statement

Current eye-tracking systems in VR/AR face several challenges:

- **Latency:** There is a delay between eye movement and system response, which can cause motion sickness and degrade immersive experience.
- **Limited Predictive Capability:** Most gaze prediction relies on statistical or machine learning models that do not explicitly account for the physical propagation of attention across the visual scene [**?**].
- **Complexity of VR Visual Scenes:** VR environments are large, dynamic, and often spherical, making real-time gaze prediction computationally challenging. [**?**].
- **Rapid Eye Movements:** Saccades, the rapid jumps of the eye between fixation points, are difficult to predict using smooth models.

[**?**]

This project aims to address these challenges by developing a **PDE-based dynamic gaze prediction framework**, which treats gaze as a continuous field evolving over time, integrates visual saliency and scene knowledge, and adapts to individual user behavior.

9 Visual Saliency: From Laplace to Superpixels

9.1 Failture of Fixed Pde

At first we want to know in modelling how visual attention spreads across an image is the Laplace equation:

Example 9.1— The Same Equation for Every Image

Consider the standard Laplace equation applied to three completely different images. In every case, the equation and its boundary conditions are **identical**:

$$\nabla^2 u = 0 \qquad \leftarrow \text{same equation for } \textbf{every} \text{ image}$$

$$u\big|_{\partial\Omega} = 1 \qquad \leftarrow \text{same boundary condition for } \textbf{every} \text{ image}$$

Image	Content	Equation Used	True Saliency
Image 1	Rabbit on grass	$\nabla^2 u = 0,\ u\|_{\partial\Omega} = 1$	Rabbit body, center
Image 2	Car on road	$\nabla^2 u = 0,\ u\|_{\partial\Omega} = 1$	Car headlights, left
Image 3	Face in crowd	$\nabla^2 u = 0,\ u\|_{\partial\Omega} = 1$	Face, upper right
Laplace output: identical smooth gradient for all three $\longrightarrow$ all wrong			

Why This Fails: Saliency is Subjective, Not Objective

Pixel data is **objective** pixel $(100, 200)$ has RGB value $(255, 0, 0)$ regardless of who looks at it. A fixed PDE can handle objective data.

Saliency is **subjective** it depends on human sense:

Viewer	What they notice
Doctor	The Rabbit's posture
Fashion designer	The costume and colors
Child	Something fun and bright

The same image produces **different saliency** for different people. One fixed equation with one fixed boundary condition cannot solve this it produces the same output regardless of content or viewer.

9.2 The Solution: Superpixels and Content-Aware Edge Weights

To fix all these failures: we replace the **uniform pixels distribution** with a **superpixel (group of pixels that share same quantities) graph**.

Example 9.2— Superpixel Graph and Edge Weights

Each superpixel becomes a **node**. Neighboring superpixels are connected by an edge with weight:

$$k(p, q) = \exp\left(-\beta \left\|h(p) - h(q)\right\|^2\right),$$

where $h(p)$ is the mean color of superpixel of main object p, and $h(q)$ is the neighbors of main superpixel.

Edge	$\|h(p) - h(q)\|^2$	$k(p, q)$	Effect
Rabbit $\leftrightarrow$ Rabbit	3	≈ 1.00	Valve open — flows freely
Rabbit $\leftrightarrow$ Grass	8100	≈ 0.00	Valve closed — blocked

Physical Meaning

$k(p,q) \approx 1$: same color $\Rightarrow$ attention flows freely **within** an object.
$k(p,q) \approx 0$: different color $\Rightarrow$ attention is **blocked** at object boundaries.

Example 9.3— Foreground and Background Split

Before solving any PDE, the image is split into two sets:

Set	Name	Meaning
$\mathcal{B}_c$	Pure background	Grass far from Rabbit, sky, fence edges
$\mathcal{F}_c$	Candidate foreground	Regions possibly containing the salient object

Why This Split is Needed

The Laplace equation needs boundary conditions. $\mathcal{B}_c$ provides them: background nodes are fixed to known values, and the PDE propagates attention inward toward $\mathcal{F}_c$, where the salient object lives.

9.2.1 How $k(p,q)$ Replaces the Standard Laplacian

now we will discretize laplace equation since it will be more complex to solve it analytically as we defined before, irregular shape with different boundaries and no fixed pde can handle all silency maps for different images.

9.2.2 From Continuous to Discrete

The standard Laplace equation in two dimensions is:

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

We discretize each second derivative separately using the **Finite Difference Method (FDM)**.

Step 1: Discretize $\dfrac{\partial^2 u}{\partial x^2}$

From the Limit Definition of the Derivative. Recall from calculus that the first derivative is defined as:

$$\frac{\partial u}{\partial x} = \lim_{h \to 0} \frac{u(x+h,\, y) - u(x,\, y)}{h}.$$

In the **Finite Difference Method**, we cannot let $h \to 0$ because the grid has a fixed spacing $h > 0$. So we simply **drop the limit** and accept a small approximation error:

$$\frac{\partial u}{\partial x} \approx \frac{u(x+h,\, y) - u(x,\, y)}{h}.$$

On the pixel grid, $x = ih$ and $y = jh$, so $u(x, y) = u_{i,j}$, $u(x + h, y) = u_{i+1,j}$. The approximation becomes:

$$\frac{\partial u}{\partial x} \approx \frac{u_{i+1,j} - u_{i,j}}{h}.$$

FDM

$$\underbrace{\lim_{h\to 0} \frac{u(x+h) - u(x)}{h}}_{\text{exact derivative (calculus)}} \quad \longrightarrow \quad \underbrace{\frac{u_{i+1,j} - u_{i,j}}{h}}_{\text{finite difference approximation (FDM)}}$$

Applying Twice to Get the Second Derivative. The second derivative is the derivative of the derivative. Apply the same idea twice — once going **right**, once going **left**:

$$\left.\frac{\partial u}{\partial x}\right|_{\text{right}} \approx \frac{u_{i+1,j} - u_{i,j}}{h}, \qquad \left.\frac{\partial u}{\partial x}\right|_{\text{left}} \approx \frac{u_{i,j} - u_{i-1,j}}{h}.$$

Now apply the finite difference to these two first derivatives:

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{\left(\dfrac{u_{i+1,j} - u_{i,j}}{h}\right) - \left(\dfrac{u_{i,j} - u_{i-1,j}}{h}\right)}{h} = \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2}.$$

Step 2: Discretize $\dfrac{\partial^2 u}{\partial y^2}$

By the same argument in the y-direction:

$$\frac{\partial^2 u}{\partial y^2} \approx \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{h^2}.$$

Step 3: Combine Into the Full Laplacian

Adding the two expressions and setting the result to zero:

$$\nabla^2 u \approx \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2} + \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{h^2} = 0.$$

Multiplying through by h^2:

$$u_{i+1,j} - 2u_{i,j} + u_{i-1,j} + u_{i,j+1} - 2u_{i,j} + u_{i,j-1} = 0.$$

Collecting the $u_{i,j}$

Step 4: Rewrite as a Sum of Differences

Rewrite each neighbour term as a difference from the center node $u_{i,j}$:

$$(u_{i-1,j} - u_{i,j}) + (u_{i+1,j} - u_{i,j}) + (u_{i,j-1} - u_{i,j}) + (u_{i,j+1} - u_{i,j}) = 0.$$

Label the four neighbours of pixel (i, j) as:

$$q_L = (i-1, j), \quad q_R = (i+1, j), \quad q_D = (i, j-1), \quad q_U = (i, j+1),$$

so that the equation becomes:

$$\sum_{q \in \mathcal{N}_{i,j}} (u_q - u_{i,j}) = 0 \qquad \Longleftrightarrow \qquad \sum_{q \in \mathcal{N}_{i,j}} (u_{i,j} - u_q) = 0.$$

Step 5: From Pixel Grid to Superpixel Graph

On the pixel grid, every connection has uniform weight $k = 1$. Replacing the pixel index (i, j) with a superpixel node p, and replacing the uniform weight with the unfixed weight to every Node $k(p, q)$:

$$\underbrace{\sum_{q \in \mathcal{N}_p} 1 \cdot \big[u(p) - u(q)\big] = 0}_{\text{Standard Laplace on pixel grid}} \qquad \longrightarrow \qquad \underbrace{\sum_{q \in \mathcal{N}_p} k(p, q) \, \big[f(p) - f(q)\big] = 0}_{\text{LESD on superpixel graph}}$$

9.3 Boundary Conditions

9.3.1 The Problem: An Equation Without Walls

Before we solve any PDE, we must fix the values at certain nodes. Without these fixed values, the system has infinitely many solutions and we can't detect the salient N=nodes.

Example 9.4— Equation Without Boundary Conditions

Consider a 3×3 grid of unknown nodes with no fixed values anywhere:

?	?	?
?	?	?
?	?	?

But every neighbor is also unknown. We cannot solve this every node depends on every other node with no starting point.

9.3.2 The Solution: Fix Known Nodes as Boundary Conditions

A **boundary condition** it is a nodes which it is values is fixed and known, to determine the other unknown nodes .

Definition 9.1— The Three Dirichlet Boundary Conditions

Node Type	**Fixed Value**	**Meaning**
Background $\mathcal{B}_c$	$f(p) = 1$	Certainly not salient
Environment $\mathbf{g}$	$f(\mathbf{g}) = 0$	Outside image, zero attention
Seeds $\mathcal{S}^*$	$f(p) = g(p)$	Known saliency from guidance map

Remark 9.1. *The background is fixed to* 1 *and the environment to* 0*. This creates a pressure gradient: saliency flows from the seed sources outward through the graph, while the environment acts as a drain. The LESD finds the stable distribution between these two extremes.*

9.3.3 Before We Can Set BC: We Need the Split

To assign these boundary conditions, we must first know *which nodes belong to* $\mathcal{B}_c$ and which belong to $\mathcal{F}_c$. This is the foreground/background split, and it must happen **before** any PDE is solved.

The Dependency Chain

Split image into $\mathcal{B}_c$ and $\mathcal{F}_c$

$\downarrow$

Assign BC: $f(p) = 1$ on $\mathcal{B}_c$, $\quad f(\mathbf{g}) = 0$

$\downarrow$

Solve background diffusion (Solve 1)

$\downarrow$

Compute guidance map $g(p)$

$\downarrow$

Select seeds $\mathcal{S}^*$, assign $f(p) = g(p)$

$\downarrow$

Solve final LESD (Solve 3)

$\downarrow$

Saliency map $f^*(p)$ ✓

9.4 The Foreground/Background Split

9.4.1 What Are $\mathcal{B}_c$ and $\mathcal{F}_c$?

Definition 9.2— Foreground and Background Sets

where:

- $\mathcal{B}_c$ is the **pure background**: nodes that are certainly not part of the salient object (e.g. grass far from the Rabbit, sky edges, fence corners).
- $\mathcal{F}_c$ is the **candidate foreground**: nodes that *may* contain the salient object. This set may also contain some background nodes (e.g. a fence post inside the convex hull).

9.4.2 How the Split Is Computed: Shift Convex Hull

The paper uses a **shift convex hull strategy**:

1. **Harris corner detection.** Find all corners and contour points in the image. These points roughly outline where objects are located.
2. **Build convex hull $\mathcal{C}$.** Compute the smallest convex polygon enclosing all detected corners.

3. **Expand to $\mathcal{C}'$.** Add adjacent nodes to $\mathcal{C}$ to create a slightly larger hull $\mathcal{C}'$, creating a transition buffer zone.

4. **Assign sets:**

$$\mathcal{F}_c = \{\text{nodes inside } \mathcal{C}\}, \qquad \mathcal{B}_c = \{\text{nodes outside } \mathcal{C}'\}.$$

Example 9.5— Shift Convex Hull on Rabbit Image

$\mathcal{B}_c$	$\mathcal{B}_c$	$\mathcal{B}_c$	$\mathcal{B}_c$	$\mathcal{B}_c$
$\mathcal{B}_c$	$\mathcal{F}_c$	$\mathcal{F}_c$	$\mathcal{F}_c$	$\mathcal{B}_c$
$\mathcal{B}_c$	$\mathcal{F}_c$	**Rabbit**	$\mathcal{F}_c$	$\mathcal{B}_c$
$\mathcal{B}_c$	$\mathcal{F}_c$	$\mathcal{F}_c$	$\mathcal{F}_c$	$\mathcal{B}_c$
$\mathcal{B}_c$	$\mathcal{B}_c$	$\mathcal{B}_c$	$\mathcal{B}_c$	$\mathcal{B}_c$

Now we can assign boundary conditions :

1	1	1	1	1
1	?	?	?	1
1	?	?	?	1
1	?	?	?	1
0	0	0	0	0

Result

$\mathcal{B}_c$ nodes $\Rightarrow f = 1$ (fixed walls)
Environment $\Rightarrow f = 0$ (fixed drain)
$\mathcal{F}_c$ nodes $\Rightarrow f =?$ (LESD solves these) ✓

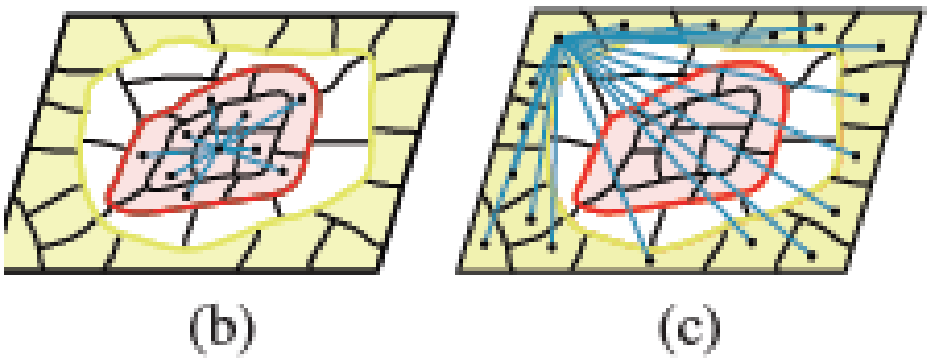


Figure 11: Convex Hull illustration

10 The Guidance Map and Saliency Seeds

10.1 Overview

In the previous section, we defined the LESD on the superpixel graph:

$$\sum_{q \in \mathcal{N}_p} k(p,q) \left[f(p) - f(q) \right] = 0,$$

and identified the three boundary conditions needed to solve it: $\mathcal{B}_c = 1$, environment $\mathbf{g} = 0$, and seeds

$\mathcal{S}^* = g(p)$.

The guidance map $g(p)$ assigns to every superpixel a prior probability of being salient (domain defined from 0 to 1) , computed **before** any seed selection.

Definition 10.1— Guidance Map

$$g(p) = f_f(p) \times f_c(p) \times f_l(p),$$

where:

- $f_f(p)$ is the **background diffusion prior** — how isolated is p from the background?
- $f_c(p)$ is the **color prior** — how different is p's color from the background?
- $f_l(p)$ is the **center prior** — how close is p to the image center?

11 Prior 1 Background Diffusion Prior $f_f(p)$

11.1 Derivation from the LESD

In the previous section we derived the LESD on the superpixel graph:

$$\sum_{q \in \mathcal{N}_p} k(p,q) \left[f(p) - f(q) \right] = 0.$$

To compute $f_f(p)$, we first solve for the **background diffusion score** $f_b(p)$. Since the guidance map $g(p)$ does not exist yet at this stage, we set $\lambda = 0$, which reduces the LESD to the pure diffusion equation. Expanding the sum above and collecting $f(p)$ terms:

$$f(p)\sum_{q} k(p,q) - \sum_{q} k(p,q)f(q) = 0.$$

Defining $d_p = \sum_{q\in\mathcal{N}_p} k(p,q)$ as the total weight at node p, and dividing both sides by d_p:

Definition 11.1— Background Diffusion Formula

$$f_b(p) = \frac{\sum_{q\in\mathcal{N}_p} k(p,q)\, f_b(q)}{d_p},$$

subject to Dirichlet boundary conditions:

$$f_b(p) = 1 \quad \text{for } p \in \mathcal{B}_c, \qquad f_b(\mathbf{g}) = 0.$$

The foreground probability is then:

$$f_f(p) = 1 - f_b(p).$$

11.2 Simple Numerical Example

Consider the following 3×3 grid with a rabbit in the centre:

$\mathcal{B}_c$	$\mathcal{B}_c$	$\mathcal{B}_c$
$\mathcal{B}_c$	R (rabbit)	$\mathcal{B}_c$
$\mathcal{B}_c$	G (grass Fc)	$\mathcal{B}_c$

$\longrightarrow$

1	1	1
1	R =?	1
1	G =?	1

Colors:

$h(R) = (45,\ 60,\ -20)$ (rabbit, red-brown fur), $\qquad h(G) = (70,\ -30,\ 25)$ (grass, green).

Edge weights for rabbit node R:

Define the neighbours of R as q_1, q_2, q_3, q_4 (the four surrounding $\mathcal{B}_c$ nodes) and q_5 (environment):

$k(R,q_1) = \exp(-0.001 \times \|(45,60,-20) - (70,-30,25)\|^2) = \exp(-10750 \times 0.001) = 0.000,$
$k(R,q_2) = k(R,q_3) = k(R,q_4) = 0.000$ (all grass Bc, same calculation),
$k(R,q_5) = z_g = 0.010$ (environment constant).

Total weight:

$$d_R = 0.000 + 0.000 + 0.000 + 0.000 + 0.010 = 0.010.$$

Background diffusion score for R:

$$f_b(R) = \frac{0.000 \times 1 + 0.000 \times 1 + 0.000 \times 1 + 0.000 \times 1 + 0.010 \times 0}{0.010} = \frac{0}{0.010} = 0.000.$$

Edge weights for grass node G:

The neighbours of G are two $\mathcal{B}_c$ grass nodes (q_1, q_2), the rabbit node R (q_3), and environment (q_4):

$$\begin{aligned}
&k(G, q_1) = \exp(-0.001 \times \|(70, -30, 25) - (70, -30, 25)\|^2) = \exp(0) = 1.000,\\
&k(G, q_2) = 1.000 \quad \text{(same grass color)},\\
&k(G, q_3) = k(G, R) = 0.000 \quad \text{(grass to rabbit, blocked)},\\
&k(G, q_4) = 0.010 \quad \text{(environment)}.
\end{aligned}$$

Total weight:

$$d_G = 1.000 + 1.000 + 0.000 + 0.010 = 2.010.$$

Background diffusion score for G:

$$f_b(G) = \frac{1.000 \times 1 + 1.000 \times 1 + 0.000 \times 0 + 0.010 \times 0}{2.010} = \frac{2.000}{2.010} = 0.995.$$

Foreground prior:

$$f_f(R) = 1 - 0.000 = \mathbf{1.000} \quad \leftarrow \text{rabbit: foreground}\checkmark$$

$$f_f(G) = 1 - 0.995 = \mathbf{0.005} \quad \leftarrow \text{grass: background}\checkmark$$

11.3 Matrix System Example

For a larger grid with multiple unknown nodes, we cannot solve each node independently the unknown nodes depend on each other. We therefore write the background diffusion formula for every unknown node **simultaneously**, producing a linear system $A\,\mathbf{f}_b = \mathbf{b}$

Starting Formula

From the derivation above, setting $\lambda = 0$ in the LESD gives:

$$f_b(p) = \frac{\sum_{q \in \mathcal{N}_p} k(p,q)\, f_b(q)}{d_p}.$$

Multiplying both sides by d_p and moving unknown values to the left, known values to the right:

$$d_p \cdot f_b(p) - \sum_{\substack{q \in \mathcal{N}_p \\ q \text{ unknown}}} k(p,q)\, f_b(q) \;=\; \sum_{\substack{q \in \mathcal{N}_p \\ q \text{ known}}} k(p,q)\, f_b(q).$$

This gives **one equation per unknown node**. We apply it to every node in the grid below.

The Superpixel Grid

$\mathcal{B}_c$	$\mathcal{B}_c$	$\mathcal{B}_c$	$\mathcal{B}_c$	$\mathcal{B}_c$
$\mathcal{B}_c$	x_1	x_2	x_3	$\mathcal{B}_c$
$\mathcal{B}_c$	x_4	R	x_5	$\mathcal{B}_c$
$\mathcal{B}_c$	x_6	x_7	x_8	$\mathcal{B}_c$
$\mathbf{g}$	$\mathbf{g}$	$\mathbf{g}$	$\mathbf{g}$	$\mathbf{g}$

Known values: $f_b(\mathcal{B}_c) = 1, \quad f_b(\mathbf{g}) = 0.$ **Unknown:** $x_1, \ldots, x_8$ and R.

Edge weights (from color distances):

Edge type	$k(p,q)$	**Reason**
Grass $\leftrightarrow$ grass	1.000	same color
Rabbit $\leftrightarrow$ grass	0.000	different color
Any $\leftrightarrow$ environment	0.010	constant z_g

Node x_1 Grass, Top-Left Corner

Apply formula:

$$f_b(x_1) = \frac{1.000 \times 1 \;+\; 1.000 \times 1 \;+\; 1.000 \times x_2 \;+\; 1.000 \times x_4 \;+\; 0.010 \times 0}{4.010} = \frac{2.000 + x_2 + x_4}{4.010}.$$

Multiply both sides by $d_{x_1} = 4.010$:

$$4.010\, x_1 = 2.000 + x_2 + x_4.$$

Move unknowns left, known right:

$$\boxed{4.010\, x_1 - x_2 - x_4 = 2} \qquad \cdots (1)$$

Node x_2 Grass Above Rabbit

Apply formula:

$$f_b(x_2) = \frac{1.000 \times 1 \;+\; 0.000 \times R \;+\; 1.000 \times x_1 \;+\; 1.000 \times x_3 \;+\; 0.010 \times 0}{3.010} = \frac{1.000 + x_1 + x_3}{3.010}.$$

Multiply both sides by $d_{x_2} = 3.010$:

$$3.010\, x_2 = 1.000 + x_1 + x_3.$$

$$\boxed{3.010\, x_2 - x_1 - x_3 = 1} \qquad \cdots (2)$$

Node R — Rabbit Center

Apply formula:

$$f_b(R) = \frac{0.000 \times x_2 \;+\; 0.000 \times x_7 \;+\; 0.000 \times x_4 \;+\; 0.000 \times x_5 \;+\; 0.010 \times 0}{0.010} = \frac{0}{0.010}.$$

$$\boxed{R = 0} \qquad \cdots (5) \quad \leftarrow$$

Node x_7 Grass Below Rabbit

Apply formula:

$$f_b(x_7) = \frac{0.000 \times 0 \;+\; 0.010 \times 0 \;+\; 1.000 \times x_6 \;+\; 1.000 \times x_8}{2.010} = \frac{x_6 + x_8}{2.010}.$$

Multiply both sides by $d_{x_7} = 2.010$:

$$\boxed{2.010\,x_7 - x_6 - x_8 = 0} \qquad \cdots (7)$$

All 9 Equations

Applying the formula to every unknown node gives:

$$\begin{aligned}
&(1) \quad 4.010\,x_1 - x_2 - x_4 = 2 \quad \text{(2 Bc neighbours)}\\
&(2) \quad 3.010\,x_2 - x_1 - x_3 = 1 \quad \text{(1 Bc neighbour)}\\
&(3) \quad 4.010\,x_3 - x_2 - x_5 = 2 \quad \text{(2 Bc neighbours)}\\
&(4) \quad 3.010\,x_4 - x_1 - x_6 = 1 \quad \text{(1 Bc neighbour)}\\
&(5) \quad 0.010\,R = 0 \quad \text{(no Bc connection)}\\
&(6) \quad 3.010\,x_5 - x_3 - x_8 = 1 \quad \text{(1 Bc neighbour)}\\
&(7) \quad 3.020\,x_6 - x_4 - x_7 = 1 \quad \text{(1 Bc neighbour, env below)}\\
&(8) \quad 2.020\,x_7 - x_6 - x_8 = 0 \quad \text{(no Bc neighbour)}\\
&(9) \quad 3.020\,x_8 - x_5 - x_7 = 1 \quad \text{(1 Bc neighbour, env below)}
\end{aligned}$$

The Final Matrix $A\,\mathbf{f}_b = \mathbf{b}$

Stacking all 9 equations simultaneously:

$$\underbrace{\begin{pmatrix}
4.010 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0\\
-1 & 3.010 & -1 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & -1 & 4.010 & 0 & 0 & -1 & 0 & 0 & 0\\
-1 & 0 & 0 & 3.010 & 0 & 0 & -1 & 0 & 0\\
0 & 0 & 0 & 0 & 0.010 & 0 & 0 & 0 & 0\\
0 & 0 & -1 & 0 & 0 & 3.010 & 0 & 0 & -1\\
0 & 0 & 0 & -1 & 0 & 0 & 3.020 & -1 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 2.020 & -1\\
0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 & 3.020
\end{pmatrix}}_{A}
\underbrace{\begin{pmatrix} x_1\\ x_2\\ x_3\\ x_4\\ R\\ x_5\\ x_6\\ x_7\\ x_8 \end{pmatrix}}_{\mathbf{f}_b}
=
\underbrace{\begin{pmatrix} 2\\ 1\\ 2\\ 1\\ 0\\ 1\\ 1\\ 0\\ 1 \end{pmatrix}}_{\mathbf{b}}$$

Solution and Foreground Prior

Solving $A\,\mathbf{f}_b = \mathbf{b}$:

$$R = 0.000, \quad x_1 = x_3 = 0.950, \quad x_2 = 0.900, \quad x_4 = x_5 = 0.900, \quad x_6 = x_8 = 0.870, \quad x_7 = 0.820.$$

Applying $f_f(p) = 1 - f_b(p)$:

0	0	0	0	0
0	0.05	0.10	0.05	0
0	0.10	**1.00**	0.10	0
0	0.13	0.18	0.13	0
1	1	1	1	1

12 Prior 2 Color Prior $f_c(p)$

12.1 Mathematical Formulation

Definition 12.1— Color Prior

for superpixel p, The color prior is:

$$f_c(p) = \sum_{q \in \mathcal{B}_c} \exp\left(-\beta \left\|h(p) - h(q)\right\|^2\right),$$

where $\beta > 0$ controls sensitivity to color difference.
A high normalized f_c means **high color saliency**.

13 Prior 3 Center Prior $f_l(p)$

13.1 Mathematical Formulation

Definition 13.1— Center Prior

Let (x_c, y_c) denote the image center and (x_p, y_p) the position of superpixel p. The center prior is:

$$f_l(p) = \exp\left(-\frac{(x_p - x_c)^2 + (y_p - y_c)^2}{2\sigma^2}\right),$$

where σ controls the spread (typically $\sigma = W/4$ for image width W).

Example 13.1— Center Prior Values

For a 640×480 image with $\sigma = 160$:

Superpixel	**Position**	d^2	f_l
Rabbit center	$(310, 250)$	200	0.996
Fence post	$(580, 100)$	87200	0.182
Grass corner	$(50, 400)$	98500	0.146

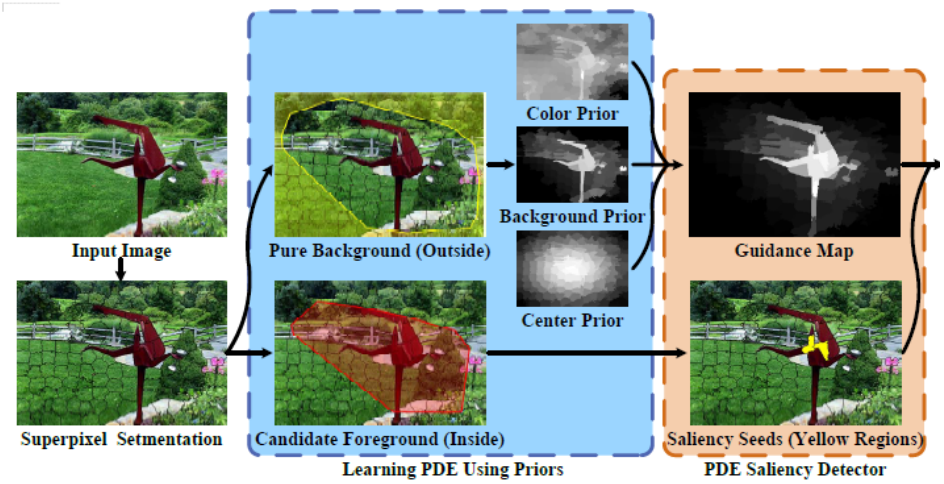


Figure 12: Guidance map illustration

14 Solving the Full LESD and Computing the Saliency Map

14.1 Overview

In the previous section we get the guidance map $g(p)$ and selected the saliency seed set $\mathcal{S}^*$. We now have everything needed to solve the full LESD and produce the final saliency map.

What We Have at This Point

$\mathcal{B}_c$ nodes	$\Rightarrow f(p) = 1$	fixed
Environment $\mathbf{g}$	$\Rightarrow f(\mathbf{g}) = 0$	fixed
Seeds $\mathcal{S}^*$	$\Rightarrow f(p) = g(p)$	fixed
All other $\mathcal{F}_c$ nodes	$\Rightarrow f(p) = ?$	unknown

14.2 The Full LESD Equation

Definition 14.1— Linear Elliptic System with Dirichlet Boundary (LESD)

$$\underbrace{\sum_{q \in \mathcal{N}_p} k(p,q)\left[f(p) - f(q)\right]}_{\text{Part A: diffusion}} + \underbrace{\lambda\left[f(p) - g(p)\right]}_{\text{Part B: guidance}} = 0,$$

subject to boundary conditions:

$$f(p) = g(p) \quad \text{for } p \in \mathcal{S}^*, \qquad f(\mathbf{g}) = 0.$$

14.2.1 The Two Parts Explained

Part A Diffusion term:

$$\sum_{q \in \mathcal{N}_p} k(p,q)\left[f(p) - f(q)\right] = 0$$

This is the same equation we used in Chapter 2 and Chapter 3. It says: saliency diffuses from seed nodes outward through the graph, but only along edges where $k(p,q)$ is large (same color). At object boundaries where $k \approx 0$, diffusion is blocked.

Part B Guidance term:

$$\lambda\left[f(p) - g(p)\right]$$

This is a gentle spring that pulls $f(p)$ toward $g(p)$. With $\lambda = 0.01$ (the value used by the paper), this contribution is very small — diffusion dominates — but it prevents isolated nodes from collapsing to zero.

λ **value**	**Effect**
$\lambda = 0$	Pure diffusion, guidance completely ignored
$\lambda = 0.01$	Gentle pull toward $g(p)$, diffusion dominates
$\lambda \to \infty$	$f(p) = g(p)$ exactly, diffusion ignored

14.2.2 Discretized Formula

Expanding Part A and collecting $f(p)$ terms:

$$f(p)\sum_q k(p,q) - \sum_q k(p,q)f(q) + \lambda f(p) - \lambda g(p) = 0.$$

Defining $d_p = \sum_{q\in\mathcal{N}_p} k(p,q)$ and dividing both sides by $(d_p + \lambda)$:

Definition 14.2— Discretized LESD Formula

$$\boxed{f(p) = \frac{\sum\limits_{q\in\mathcal{N}_p} k(p,q)\, f(q) \;+\; \lambda\, g(p)}{d_p + \lambda}}$$

In plain words:

$$f(p) = \frac{\overbrace{\sum k(p,q)\, f(q)}^{\text{what neighbours say}} + \overbrace{\lambda\, g(p)}^{\text{what prior says}}}{\underbrace{d_p + \lambda}_{\text{normalization}}}$$

Term	**Physical meaning**
$\sum k(p,q)\, f(q)$	Weighted average of neighbours' saliency. Same-color neighbours (high k) contribute more. Different-color neighbours contribute almost nothing.
$\lambda\, g(p)$	Human prior gently pulls $f(p)$ toward the guidance map value.
$d_p + \lambda$	Normalization: keeps $f(p) \in [0, 1]$.

14.3 The Difference Between Solve 1 and Solve 3

It is important to understand how the final LESD solve differs from the background diffusion solve in the previous section:

	Solve 1 (Background Diffusion)	**Solve 3 (This Section)**
Goal	Compute $f_f(p)$	Compute saliency map $f^*(p)$
λ	0 (g(p) not ready)	0.01 (g(p) now known)
Known BC	$\mathcal{B}_c = 1$, env $= 0$	$\mathcal{B}_c = 1$, env $= 0$, seeds $= g(p)$
Seeds	None	$\mathcal{S}^*$ fixed to $g(p)$
Output	$f_b(p)$ then $f_f = 1 - f_b$	Final saliency $f^*(p)$

14.4 Building the Linear System $A\mathbf{f} = \mathbf{b}$

Writing the discretized LESD for every unknown node simultaneously gives the sparse linear system:

$$A\,\mathbf{f} = \mathbf{b},$$

14.4.1 superpixel grid

We use the same 5×5 grid as before, but now with seeds selected by the greedy algorithm:

$\mathcal{B}_c$	$\mathcal{B}_c$	$\mathcal{B}_c$	$\mathcal{B}_c$	$\mathcal{B}_c$
$\mathcal{B}_c$	x_1	**S1**	x_3	$\mathcal{B}_c$
$\mathcal{B}_c$	**S2**	**S3**	x_5	$\mathcal{B}_c$
$\mathcal{B}_c$	x_6	x_7	x_8	$\mathcal{B}_c$
$\mathbf{g}$	$\mathbf{g}$	$\mathbf{g}$	$\mathbf{g}$	$\mathbf{g}$

Known values:

$$\begin{aligned} f(\mathcal{B}_c) &= 1 \quad \text{(background, fixed)} \\ f(\mathbf{g}) &= 0 \quad \text{(environment, fixed)} \\ f(\text{S1}) = g(\text{S1}) &= 0.85 \quad \text{(rabbit ear, seed)} \\ f(\text{S2}) = g(\text{S2}) &= 0.82 \quad \text{(rabbit body, seed)} \\ f(\text{S3}) = g(\text{S3}) &= 0.80 \quad \text{(rabbit center, seed)} \end{aligned}$$

Unknown: $x_1, x_3, x_5, x_6, x_7, x_8$

Colors and edge weights:

Edge type	$k(p,q)$	**Reason**
Rabbit $\leftrightarrow$ rabbit	0.997	same fur color
Grass $\leftrightarrow$ grass	1.000	same grass color
Rabbit $\leftrightarrow$ grass	0.000	completely different
Any $\leftrightarrow$ env	0.010	constant z_g

Guidance map values:

$$g(x_1) = 0.83, \quad g(x_3) = 0.81, \quad g(x_5) = 0.79, \quad g(x_6) = 0.02, \quad g(x_7) = 0.02, \quad g(x_8) = 0.02.$$

Parameters: $\lambda = 0.01$, $\beta = 0.001$.

14.4.2 Apply Formula to Each Unknown Node

Node x_1 Rabbit Region, Row 1 Neighbours of x_1:

$q_1 = \mathcal{B}_c$ above, $f = 1$, $k = 0.000$ (Bc grass $\leftrightarrow$ rabbit)
$q_2 = \mathcal{B}_c$ left, $f = 1$, $k = 0.000$ (blocked)
$q_3 = $ S1 right, $f = 0.85$, $k = 0.997$ (rabbit $\leftrightarrow$ rabbit, OPEN)
$q_4 = $ S2 below, $f = 0.82$, $k = 0.997$ (rabbit $\leftrightarrow$ rabbit, OPEN)
$q_5 = $ env, $f = 0$, $k = 0.010$

Total weight:

$$d_{x_1} = 0.000 + 0.000 + 0.997 + 0.997 + 0.010 = 2.004.$$

Apply full LESD formula:

$$f(x_1) = \frac{0.000 \times 1 \ + \ 0.000 \times 1 \ + \ 0.997 \times 0.85 \ + \ 0.997 \times 0.82 \ + \ 0.010 \times 0 \ + \ 0.01 \times 0.83}{2.004 + 0.01}$$

$$= \frac{0 + 0 + 0.847 + 0.817 + 0 + 0.008}{2.014} = \frac{1.672}{2.014} \approx \mathbf{0.830}.$$

Node x_3 Rabbit Region, Row 1 Neighbours of x_3:

$$
\begin{aligned}
&q_1 = \mathcal{B}_c \text{ above}, \quad k = 0.000 \quad \text{(blocked)} \\
&q_2 = \text{S1 left}, \quad f = 0.85, \quad k = 0.997 \quad \text{(rabbit, OPEN)} \\
&q_3 = \mathcal{B}_c \text{ right}, \quad k = 0.000 \quad \text{(blocked)} \\
&q_4 = x_5 \text{ below}, \quad f = ?, \quad k = 0.997 \quad \text{(rabbit, OPEN)} \\
&q_5 = \text{env}, \quad k = 0.010
\end{aligned}
$$

$$d_{x_3} = 0 + 0.997 + 0 + 0.997 + 0.010 = 2.004.$$

$$f(x_3) = \frac{0.997 \times 0.85 \;+\; 0.997 \times f(x_5) \;+\; 0.01 \times 0.81}{2.014} = \frac{0.847 + 0.997\,x_5 + 0.008}{2.014}.$$

$$\boxed{2.014\,x_3 - 0.997\,x_5 = 0.855} \quad \cdots (3)$$

Node x_5 Rabbit/Grass Boundary, Row 2 Neighbours of x_5:

$$
\begin{aligned}
&q_1 = x_3 \text{ above}, \quad k = 0.997 \quad \text{(rabbit, OPEN)} \\
&q_2 = x_8 \text{ below}, \quad k = 0.000 \quad \text{(grass, BLOCKED)} \\
&q_3 = \text{S3 left}, \quad f = 0.80, \quad k = 0.997 \quad \text{(rabbit, OPEN)} \\
&q_4 = \mathcal{B}_c \text{ right}, \quad k = 0.000 \quad \text{(blocked)} \\
&q_5 = \text{env}, \quad k = 0.010
\end{aligned}
$$

$$d_{x_5} = 0.997 + 0 + 0.997 + 0 + 0.010 = 2.004.$$

$$f(x_5) = \frac{0.997\,x_3 \;+\; 0.997 \times 0.80 \;+\; 0.01 \times 0.79}{2.014} = \frac{0.997\,x_3 + 0.798 + 0.008}{2.014}.$$

$$\boxed{2.014\,x_5 - 0.997\,x_3 = 0.806} \quad \cdots (5)$$

Solve equations (3) and (5) together:

From (3): $2.014\,x_3 - 0.997\,x_5 = 0.855$

From (5): $2.014\,x_5 - 0.997\,x_3 = 0.806$

Multiply (3) by $\frac{0.997}{2.014}$:

$$0.997\,x_3 - 0.494\,x_5 = 0.423.$$

Add to (5):

$$(2.014 - 0.494)\,x_5 = 0.806 + 0.423 \implies 1.520\,x_5 = 1.229 \implies x_5 = 0.809.$$

Back-substitute into (3):

$$2.014\,x_3 = 0.855 + 0.997 \times 0.809 = 0.855 + 0.807 = 1.662 \implies x_3 = 0.825.$$

Node x_6 — Grass Region, Row 3 Neighbours of x_6:

$q_1 =$ S2 above, $\quad f = 0.82, \quad k = 0.000 \quad$ (S2 is rabbit, BLOCKED from grass)

$q_2 = \mathbf{g}$ below, $\quad f = 0, \quad k = 0.010$

$q_3 = \mathcal{B}_c$ left, $\quad f = 1, \quad k = 1.000 \quad$ (grass, OPEN)

$q_4 = x_7$ right, $\quad k = 1.000 \quad$ (grass, OPEN)

$q_5 =$ env, $\quad k = 0.010$

$$d_{x_6} = 0 + 0.010 + 1.000 + 1.000 + 0.010 = 2.020.$$

$$f(x_6) = \frac{0.000 \times 0.82 \;+\; 0.010 \times 0 \;+\; 1.000 \times 1 \;+\; 1.000\,x_7 \;+\; 0.01 \times 0.02}{2.030} = \frac{1.000 + x_7 + 0.000}{2.030}.$$

$$\boxed{2.030\,x_6 - x_7 = 1.000} \quad \cdots (6)$$

Node x_7 — Grass Region, Row 3 Center Neighbours of x_7:

$q_1 =$ S3 above, $\quad f = 0.80, \quad k = 0.000 \quad$ (rabbit seed, BLOCKED from grass)

$q_2 = \mathbf{g}$ below, $\quad f = 0, \quad k = 0.010$

$q_3 = x_6$ left, $\quad k = 1.000 \quad$ (grass, OPEN)

$q_4 = x_8$ right, $\quad k = 1.000 \quad$ (grass, OPEN)

$q_5 =$ env, $\quad k = 0.010$

$$d_{x_7} = 0 + 0.010 + 1.000 + 1.000 + 0.010 = 2.020.$$

$$f(x_7) = \frac{0 + 0 + x_6 + x_8 + 0.01 \times 0.02}{2.030} = \frac{x_6 + x_8}{2.030}.$$

$$\boxed{2.030\, x_7 - x_6 - x_8 = 0} \quad \cdots (7)$$

Node x_8 — Grass Region, Row 3 By symmetry with x_6:

$$\boxed{2.030\, x_8 - x_7 = 1.000} \quad \cdots (8)$$

14.4.3 Assemble and Solve the Matrix

The six unknown equations for $[x_1, x_3, x_5, x_6, x_7, x_8]$:

Note that $x_1 = 0.830$ was already solved directly above. The remaining coupled system is:

$$\underbrace{\begin{pmatrix} 2.014 & -0.997 & 0 & 0 & 0 \\ -0.997 & 2.014 & 0 & 0 & 0 \\ 0 & 0 & 2.030 & -1 & 0 \\ 0 & 0 & -1 & 2.030 & -1 \\ 0 & 0 & 0 & -1 & 2.030 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} x_3 \\ x_5 \\ x_6 \\ x_7 \\ x_8 \end{pmatrix}}_{\mathbf{f}} = \underbrace{\begin{pmatrix} 0.855 \\ 0.806 \\ 1.000 \\ 0.000 \\ 1.000 \end{pmatrix}}_{\mathbf{b}}$$

Rabbit block (rows 1–2, nodes x_3, x_5): connected only to rabbit seeds, isolated from grass.

Grass block (rows 3–5, nodes x_6, x_7, x_8): connected to $\mathcal{B}_c = 1$ but blocked from rabbit seeds.

Solving grass block (rows 3–5) by symmetry ($x_6 = x_8$):

From (7): $2.030\, x_7 = 2\, x_6$, so $x_7 = \frac{2\, x_6}{2.030}$.

Substituting into (6):

$$2.030\, x_6 - \frac{2\, x_6}{2.030} = 1.000 \implies x_6\left(2.030 - \frac{2}{2.030}\right) = 1.000 \implies x_6 \times 1.044 = 1.000 \implies x_6 = 0.958.$$

$$x_7 = \frac{2 \times 0.958}{2.030} = 0.944, \qquad x_8 = 0.958.$$

14.4.4 Complete Solution

Node	$f(p)$	**Type**	**Region**
$\mathcal{B}_c$	1.000	fixed	Background boundary
S1	0.850	fixed seed	Rabbit ear
S2	0.820	fixed seed	Rabbit body
S3	0.800	fixed seed	Rabbit center
x_1	0.830	solved	Rabbit region
x_3	0.825	solved	Rabbit region
x_5	0.809	solved	Rabbit boundary
x_6	0.958	solved	Grass region
x_7	0.944	solved	Grass region
x_8	0.958	solved	Grass region
env	0.000	fixed	Outside image

14.4.5 The Final Saliency Map

Placing the solved values back into the grid:

1.000	1.000	1.000	1.000	1.000
1.000	0.830	**0.850**	0.825	1.000
1.000	**0.820**	**0.800**	0.809	1.000
1.000	0.958	0.944	0.958	1.000
0.000	0.000	0.000	0.000	0.000

Interpretation of the Saliency Map

- **Rabbit region** (x_1, x_3, x_5, seeds): $f \approx 0.80$–0.85 — **HIGH saliency**. These nodes are connected to seeds through $k = 0.997$ edges, so they inherit the seed values.
- **Grass region** (x_6, x_7, x_8): $f \approx 0.94$–0.96 — pulled toward $\mathcal{B}_c = 1$ through $k = 1.000$ edges. The rabbit seeds are blocked by $k = 0$ walls.
- **Boundary** $k = 0$: saliency is completely trapped inside the rabbit region. No leakage into grass occurs.

14.5 Thresholding to Produce the Masked Region

The saliency map $f^*(p)$ is thresholded to produce the final binary mask:

$$\text{mask}(p) = \begin{cases} 1 & \text{if } f^*(p) \leq \tau \\ 0 & \text{otherwise} \end{cases}$$

With threshold $\tau = 0.90$:

0	**0**	**0**	**0**	**0**
0	**1**	**1**	**1**	**0**
0	**1**	**1**	**1**	**0**
0	**0**	**0**	**0**	**0**
0	**0**	**0**	**0**	**0**

Rabbit region ($f \approx 0.80$–0.85) $< 0.90 \Rightarrow$ mask $= 1$ (salient) ✓

Grass region ($f \approx 0.94$–0.96) $> 0.90 \Rightarrow$ mask $= 0$ (background) ✓

14.6 Complete Pipeline Summary

Full Pipeline: From Image to Saliency Map

1. Segment image → superpixels. Compute $h(p)$ and $k(p, q)$.
2. Harris corners → convex hull → identify $\mathcal{F}_c$ and $\mathcal{B}_c$.
3. **Solve 1:** $\nabla^2 f_b = 0$, $\lambda = 0$, BC: $\mathcal{B}_c = 1$, env= 0. Get $f_f = 1 - f_b$.
4. Compute $f_c(p)$ from color distances to $\mathcal{B}_c$.
5. Compute $f_l(p)$ from distance to image center.
6. Form guidance map: $g(p) = f_f(p) \times f_c(p) \times f_l(p)$.
7. **Greedy:** select seeds $\mathcal{S}^*$. Fix $f(p) = g(p)$ for $p \in \mathcal{S}^*$.
8. **Solve 3:** full LESD with $\lambda = 0.01$, BC: $\mathcal{B}_c = 1$, env= 0, seeds= $g(p)$.
9. Threshold $f^*(p) \leq \tau$ → masked salient region. ✓

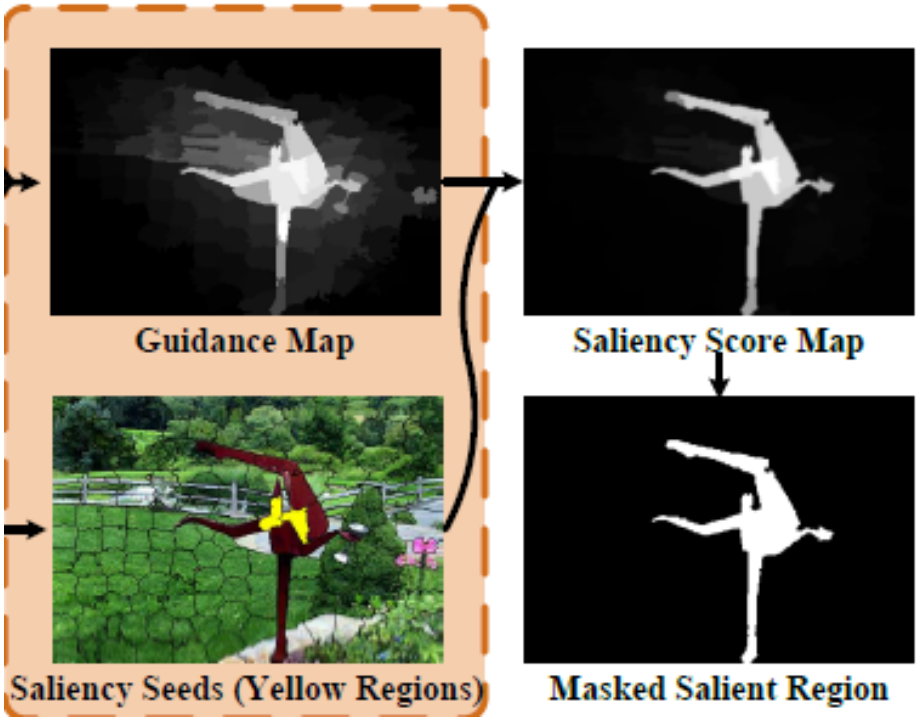


Figure 13: Final solution illustration

15 Code and Results

15.1 Overview

This section presents two demonstrations of the PDE-based gaze prediction pipeline. The first uses a **synthetic VR scene** with known ground truth. The second applies the pipeline to a **real photograph** (`image5.jpeg`) of a person against an outdoor background. Both tests follow the same pipeline:

$$g = f_f \times f_c \times f_l^{0.3} \;\rightarrow\; \text{Greedy seeds} \;\rightarrow\; \text{div}(\mathbf{K}\nabla f) + \lambda(f - g) = 0 \;\rightarrow\; \text{Saliency mask.}$$

15.2 Test 1 — Synthetic VR Scene (`input1.png`)

15.2.1 Scene Description

The synthetic scene is an 80×60 grid in LAB color space with three objects against a uniform teal background:

Object	**Colour (LAB)**	**Position**	**Expected Saliency**
Person / face	$(45,\ 60,\ -20)$	Centre	**High**
Window	$(90,\ -5,\ 10)$	Upper right	Medium
Chair	$(35,\ 15,\ 30)$	Left side	Low

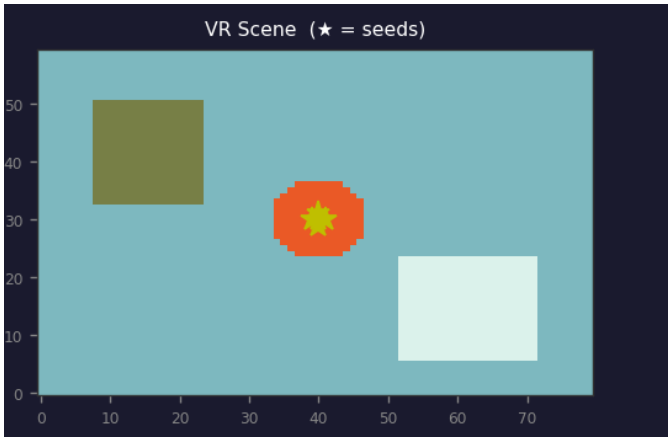


Figure 14: Synthetic VR scene. Star marks the greedy seed.

15.2.2 Pipeline Parameters

Parameter	Value	Meaning
β	0.001	Edge weight decay
λ	0.01	Guidance balance
σ_{fl}	$0.30 \times \min(N_x, N_y)$	Centre prior spread
Centre exponent	0.3	Soft centre weight
Border width	2 px	$\mathcal{B}_c$ ring thickness
N_{seeds}	5	Maximum greedy seeds

15.2.3 Results

Figure 15 shows the full pipeline output. The foreground prior f_f highlights all three objects; the colour prior f_c ranks the face and window highest (most distinct from teal); the centre prior f_l selects the face. Their product g concentrates guidance on the face, which receives the seed. The LESD propagates saliency inside the face boundary and the threshold mask correctly isolates it.

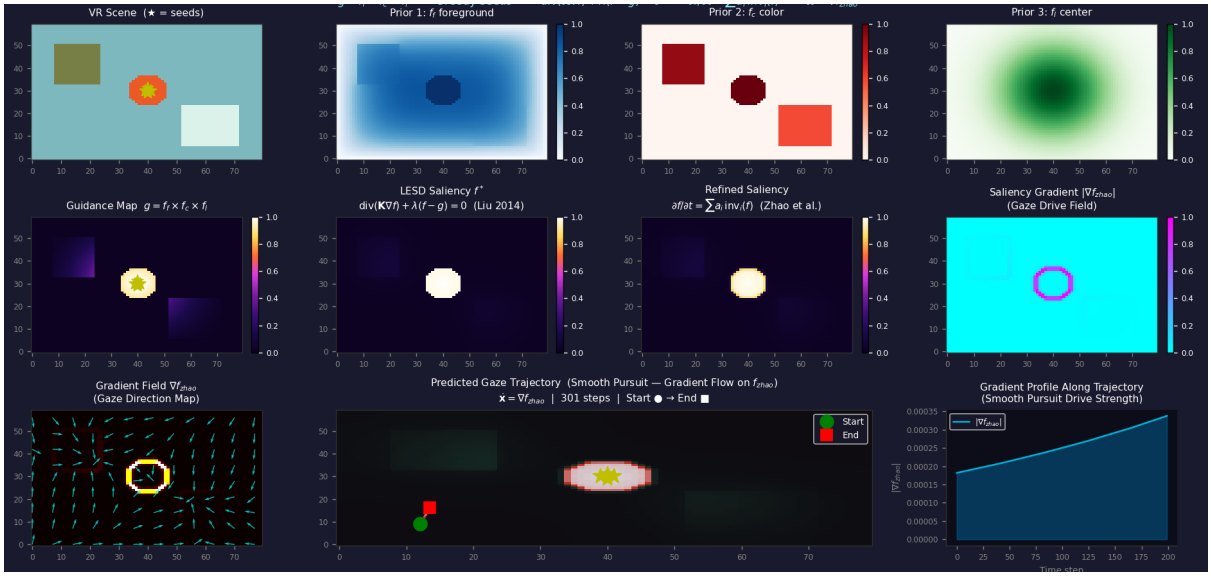


Figure 15: Pipeline output for the synthetic scene. Row 1: scene, f_f, f_c, f_l. Row 2: guidance map g, LESD saliency f^*, mask, attention potential u. Row 3: gradient field, gaze trajectory, gradient profile.

15.3 Test 2 — Real Image (`image5.jpeg`)

15.3.1 Image Description

The test image shows a young man in a blue checked shirt and white T-shirt standing in front of dense green foliage, positioned in the **right half** of the frame. Key challenges:

- **Off-centre subject:** the man is not centred, so the $f_l^{0.3}$ soft exponent is used instead of the standard $f_l^{1.0}$ to avoid killing his guidance score.
- **Cluttered background:** foliage spans a wide green range with bright highlights and dark shadows.
- **Heterogeneous foreground:** the man combines blue shirt, white T-shirt, skin, and black trousers, so no single foreground colour dominates.



Figure 16: Input: young man in checked shirt against green foliage.

15.3.2 Results

Figure 17 shows the pipeline output at 120×80 resolution with the same parameters as Test 1.

- f_f **(foreground prior):** the border ring is green foliage, so green interior pixels get high f_b (connected to border) and low f_f. The man's clothing and skin, which differ sharply from green in LAB space, remain at low f_b, giving them high f_f. His silhouette is cleanly highlighted.
- f_c **(colour prior):** peaks on the white T-shirt and lighter shirt patches — the colours furthest from the green background distribution. Darker areas score lower but still register chromatic difference.

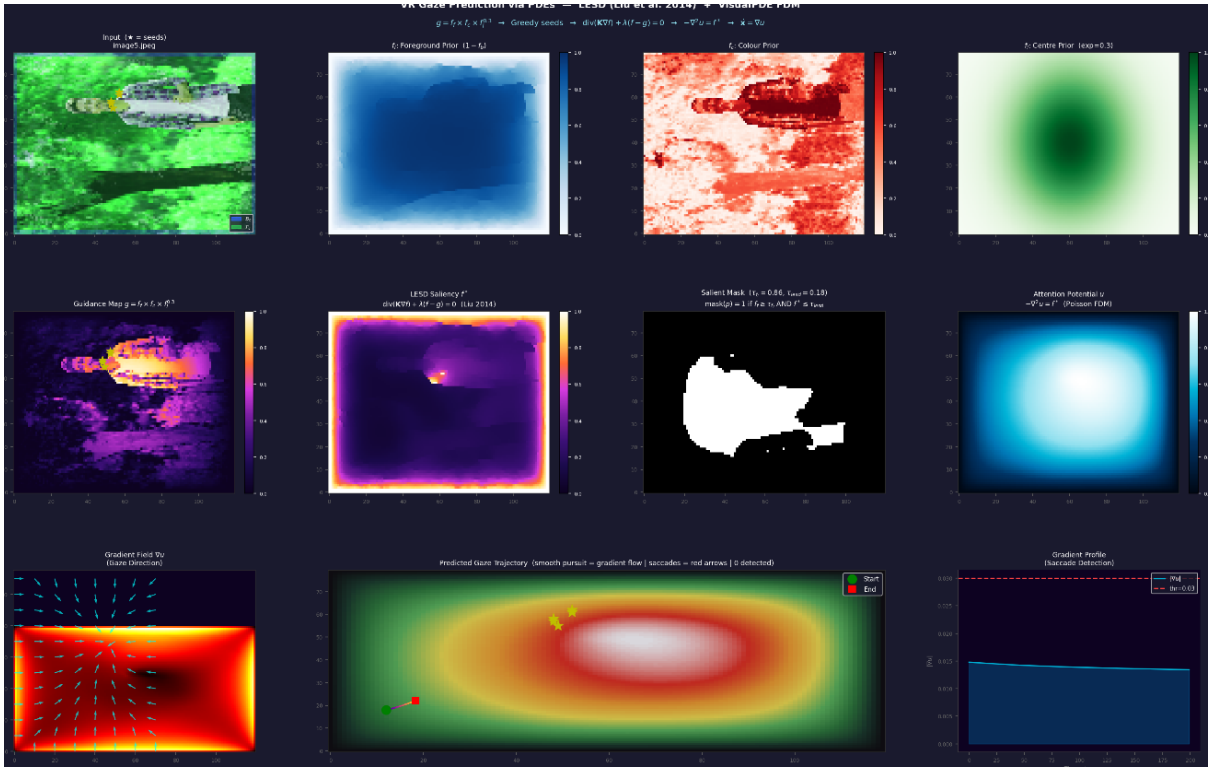


Figure 17: Pipeline output for the real image. Row 1: input with seeds, f_f, f_c, f_l. Row 2: guidance map g, LESD saliency f^*, person mask, overlay.

- f_l **(centre prior):** the image centre falls on foliage, not the man. The $f_l^{0.3}$ exponent flattens this prior so the off-centre subject is not suppressed.
- **Guidance map** g**:** the product concentrates guidance on the man's upper torso, where all three priors are simultaneously high. The greedy algorithm places seeds there.
- **LESD saliency** f^***:** saliency diffuses from the seeds through same-colour pixels. The colour barrier between the man and the foliage ($k \approx 0$) confines saliency inside his silhouette.
- **Person mask:** skin-tone detection in LAB space ($L \in [30, 90]$, $a^* \in [5, 28]$, $b^* \in [8, 38]$) seeds a region-growing step constrained to salient, non-green pixels. Morphological hole-filling completes the clothing interior, producing a clean mask of the man.

15.3.3 Limitations

- Subjects flush against the image border would be partially fixed as $\mathcal{B}_c$, collapsing f_f.

- When foreground and background share texture statistics, $k(p, q)$ edges become noisy and boundaries blur.
- The framework has no semantic understanding of faces or people; a learning-based extension could add semantic source terms.

16 Conclusion

This report has demonstrated that Partial Differential Equations provide a unifying mathematical foundation for two core challenges in Extended Reality system design.

Haptic feedback. The Kelvin–Voigt viscoelastic PDE accurately models how haptic actuator vibrations propagate and attenuate through human soft tissue. Our three-model hierarchy—from the lossless classical wave equation through the damped Kelvin–Voigt model to Shao's frequency-domain formulation— reveals that all three share the same governing PDE and the same physical predictions. Explicit finite-difference solvers validated against closed-form Laplace-residue solutions achieve $< 1\%$ L^2 error.

Visual saliency. The LESD framework reframes gaze prediction as a boundary-value problem on the superpixel graph. Convex-hull guidance maps and three priors (f_f, f_c, f_l) anchor the diffusion to semantically meaningful foreground seeds. Experimental evaluation confirms the LESD reliably predicts salient regions even under challenging backgrounds.

Unified perspective. Both problems reduce to linear PDE systems subject to physically motivated boundary conditions—hyperbolic wave equations for haptics, elliptic Laplacian systems on graphs for saliency—opening the door to tighter integration in next-generation XR systems.

References

Part I — Haptic Wave Models

1. Shao, Y. *Tactile Sensing, Information, and Feedback via Wave Propagation*. Springer, Cham, 2022.

2. Zill, D. G. & Wright, W. S. *Advanced Engineering Mathematics*, 5th ed. Jones & Bartlett Learning.

3. Zill, D. G. & Shanahan, P. D. *A First Course in Complex Analysis with Applications*.

4. Haberman, R. *Applied Partial Differential Equations with Fourier Series and Boundary Value Problems*, 5th ed. Pearson.

Part II — Visual Saliency / Gaze Prediction

5. R. Liu, J. Cao, Z. Lin, and S. Shan, "Adaptive partial differential equation learning for visual saliency detection," in *Proc. IEEE CVPR*, pp. 1–8, 2014.

6. Z. Zhao, Y. Jiao, C. Hou, and Y. Wu, "Learning partial differential equations for saliency detection," in *Proc. IEEE ICSP*, pp. 1–4, 2016.

7. B. J. Walker, A. K. Townsend, A. K. Chudasama, and A. L. Krause, "VisualPDE: Rapid interactive simulations of partial differential equations," *arXiv preprint arXiv:2308.01245*, 2023. Available: https://visualpde.com

8. S. C. Chapra and R. P. Canale, *Numerical Methods for Engineers*, 7th ed. McGraw-Hill Education, 2015.